Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments

Conjecture proofs, canonical reductions, spectral derivatives, and a source-level research audit

Research continuation prepared for Vladimir Reshetnikov and the ProveIt project

11 October 2026

Abstract

This article continues the current ProveIt polylogarithm manuscript and the six incoming reports at the pinned source revision specified below. We prove the optimal consecutive-moment conjecture for every pair of adjacent even harmonic-tail orders, with an explicit nonzero rational determinant. An elementary implementation of established bicolored parity proves the recorded Gaussian Family–2 identity and supplies an all-weight family of ordinary logarithmic integrals. For the Gamma differential hierarchy we derive all polygamma-square coefficients and a closed generating function for its complete digamma–trigamma sector. A residue-cancellation theorem converts every differential polynomial containing a polygamma derivative into an absolutely convergent Mittag–Leffler expansion, a normalized antiderivative, and convergent harmonic-zeta derivative coordinates. These tools give an explicit sixth digamma moment and constructive higher-weight formulas. We also evaluate all integer-order quadratic Hurwitz finite parts and their spectral jets, and thereby all products of positive argument derivatives of generalized Stieltjes constants, keeping the original endpoint coordinate fixed. Classical antecedents, inherited results, new proofs, unresolved arithmetic reductions, and reproducibility evidence are distinguished throughout. The remaining research questions emphasize exact identities and proof mechanisms rather than bounds.

Source baseline. `VladimirReshetnikov/ProveIt`.
Commit `f7e35be357654198c1546fd40f363694c5bcc2cc`. The audit covers the supplied manuscript and all six incoming ZIP inventories, with focused proof checks described in Section 8. The accompanying archive contains source, exact algorithms, completed verification records, and integration notes. No conjectured transcendence or independence is used in a proof.

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1 Scope, conventions, and the proof ledger

The source corpus already contains substantial all-order mathematics. In particular, the incoming reports [2, 3] establish Gamma-quotient harmonic generators, arbitrary digamma-power algorithms, the fifth moment, continuous-depth Lerch constructions, and several Stieltjes collision formulas. Repeating those results as new would obscure the actual progress. We use them as a baseline and concentrate on four concrete tasks: resolving a recorded conjecture on an infinite family, proving a recorded numerical polylogarithm relation, finding closed coefficient formulas inside the higher Gamma hierarchy, and completing explicit higher-moment and spectral-derivative calculations.

1.1 What is proved here, and what is inherited

Topic	Precise status in this continuation
Gaussian Family–2	The numerical source equation is proved exactly, and reduced to $\zeta(3)$, $\pi\beta(2)$, and $\pi^2 \log 2$. The general parity theorem is established prior work.
Adjacent even tails	The first-q conjecture is proved for all $p = q - 2 \geq 2$. The exact determinant and reconstruction identities are new developments relative to the inspected reports.
Differential normal form	The functional-monomial decomposition is classical. Its Gamma coefficient sectors and explicit endpoint specialization are developed and proved here.
Quadratic Hurwitz products	Explicit integer finite parts and every order-derivative jet follow from the classical product kernel, with the local coordinate correction proved.
Sixth digamma moment	A complete explicit formula is supplied in height-one Laurent slopes and one specified, convergent depth-at-most-three zeta-derivative combination. Its further arithmetic reduction remains open.
S_6, revised S_8, larger gaps	These remain conjectural or open. Neither the parity argument nor finite numerical evidence resolves them.

The distinction between an *explicit coordinate formula* and a *reduction to a prescribed constant alphabet* matters. Defining a constant by an unexplained divergent sum would not complete a moment calculation. Our remaining mixed coordinate is instead given by an absolutely convergent logarithmic series, a finite multiple-zeta expression, an exact value at spectral order zero, and independently checked analytic formulas. We do not assert that it is irreducible or independent of the other constants.

1.2 Harmonic, polylogarithmic, and Laurent conventions

For $n \geq 0$ and $r \geq 1$,

$$H_n^{(r)} = \sum_{k=1}^{n} k^{-r}, \qquad H_n = H_n^{(1)}, \qquad H_0^{(r)} = 0.$$

For a word $\boldsymbol{a} = (a_1, \ldots, a_d)$ of positive integers,

$$H_n(\boldsymbol{a}) = \sum_{n \geq k_1 > \cdots > k_d \geq 1} \frac{1}{k_1^{a_1} \cdots k_d^{a_d}}, \qquad H_n(\varnothing) = 1.$$

All multiple zeta functions and multiple polylogarithms in this article use *decreasing* indices:

$$\zeta(s, \boldsymbol{a}) = \sum_{n \geq 1} \frac{H_{n-1}(\boldsymbol{a})}{n^s}, \qquad \mathrm{Li}_{p,q}(x, y) = \sum_{n > m \geq 1} \frac{x^n y^m}{n^p m^q}.$$

Outside their initial domains, multiple zeta functions mean their meromorphic continuations. Where a displayed finite combination is holomorphic although individual terms have poles, the combination is formed *before* its Taylor coefficient is taken.

We use the conventional generalized Stieltjes constants

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{n\ge 0}\frac{(-1)^n\gamma_n(a)}{n!}u^n,\qquad \gamma_0(a)=-\psi(a),\quad \gamma_n=\gamma_n(1). \tag{1.1}$$

Thus the coefficient of u at $a=1$ is $-\gamma_1$, not γ_1. A second family used below is

$$\lambda_d=[u^1]\zeta(1+u,\{1\}^{d-1}),\qquad d\ge 1. \tag{1.2}$$

Here $[u^1]$ means the Laurent coefficient; the function may have a pole of order d. In particular $\lambda_1=-\gamma_1$. The higher λ_d are strict-depth Laurent slopes, not a relabeling of the ordinary γ_d. The local notation λ_2 for a dilogarithm imaginary part in the source Family–2 comparison is explicitly confined to that comparison.

The basic differential identities are

$$\partial_a\zeta(s,a)=-s\zeta(s+1,a),\qquad \psi^{(r)}(a)=(-1)^{r+1}r!\zeta(r+1,a)\quad (r\ge 1). \tag{1.3}$$

They follow directly from the defining series in its absolute convergence domain and then by continuation; see also DLMF 25.11 [9]. They fix the signs when translating the harmonic-tail identities into polygamma products.

For later comparison with antiderivatives, coefficient extraction in (1.3) gives the classical formulas

$$\gamma_n'(a)=(-1)^{n+1}\left\{\partial_s^n\zeta(s,a)\big|_{s=2}+n\,\partial_s^{n-1}\zeta(s,a)\Big|_{s=2}\right\}, \tag{1.4}$$

$$\int_{a_0}^{a}\gamma_n(v)\,\mathrm{d}v=\frac{(-1)^{n+1}}{n+1}\,\partial_s^{n+1}\{\zeta(s,a)-\zeta(s,a_0)\}\Big|_{s=0}. \tag{1.5}$$

The second term in braces in (1.4) is absent for $n=0$. The derivative formula comes from $\partial_a\zeta(1+u,a)=-(1+u)\zeta(2+u,a)$. For the primitive use $\partial_a\zeta(u,a)=-u\zeta(1+u,a)$, then differentiate $n+1$ times at zero. These formulas are included to connect the new spectral-jet identities with the conventional Stieltjes normalization; no priority claim is made for them.

1.3 The endpoint coordinate is part of the definition

If a function on $(0,1]$ has a finite principal Laurent part at zero, we define

$$\mathrm{FP}\int_0^1 f(x)\,\mathrm{d}x=\mathrm{CT}_{\epsilon=0}\int_\epsilon^1 f(x)\,\mathrm{d}x. \tag{1.6}$$

The same notation applies to finite sums of powers times logarithms, followed by an integrable remainder. The constant coefficient is taken after all negative powers and all divergent logarithms are removed. The cutoff is always the displayed coordinate $x=\epsilon$. In particular,

$$\mathrm{FP}\int_0^1 x^{-j}\,\mathrm{d}x=-\frac{1}{j-1}\ (j\ge 2),\qquad \mathrm{FP}\int_0^1 x^{-1}(\log x)^a\,\mathrm{d}x=0\ (a\ge 0).$$

For h meromorphic near $[0,1]$ with no poles on $(0,1]$,

$$\mathrm{FP}\int_0^1 h'(x)\,\mathrm{d}x=h(1)-[x^0]h(x). \tag{1.7}$$

This is an exact coefficient identity obtained by integrating from ϵ to 1. It does not replace $[x^0]h$ by the generally nonexistent value $h(0)$.

Finally, throughout the tail-rank and differential-polynomial sections, "rank" and "dimension" refer to coefficient spaces over $\mathbb{Q}$. For example, using distinct formal symbols for $\zeta(0), \zeta(2), \ldots$ does not assert their algebraic independence. Likewise the image of a derivation is a vector subspace, generally not an ideal.

2 A bilateral proof of bicolored parity and the Gaussian Family–2 identity

The canonical manuscript's equation `double:eq:fam2-example` was supported by integer-relation discovery and numerical evaluation. We give a complete proof, and simplify its right-hand side to three single-value products. The proof is a concrete application of known parity theory: Nakamura's unit-circle functional relation [4, Proposition 1.2] precedes Panzer's general multiple-polylogarithm parity theorem [5, Theorem 1.3 and (3.2)]. Our contribution here is the explicit formula in the manuscript's *decreasing-index* convention, a direct bilateral proof including indices equal to one and the collision $xy = 1$, and its exact application to the recorded Gaussian relation. It is not a claim of priority for unit-circle parity or for reducibility of all the constants displayed below.

2.1 Conventions, ordinary convergence, and the finite formula

Write

$$Z_{p,q}(x,y) = \mathrm{Li}_{p,q}(x,y) = \sum_{n>m\geq 1} \frac{x^n y^m}{n^p m^q}, \qquad p,q \geq 1.$$

In this section $|x| = |y| = 1$ and $x, y \neq 1$; the product $r = xy$ is allowed to equal one. These hypotheses include the Gaussian pairs (i,i), $(-1,i)$, $(i,-1)$, and $(i,-i)$. The ordinary nested series converges, including when $p = 1$. Indeed, for $y \neq 1$, the inner partial sums $H^{(q)}_{n-1}(y) = \sum_{m<n} y^m/m^q$ are bounded; the sequence $H^{(q)}_{n-1}(y)/n^p$ tends to zero and its successive differences are absolutely summable. Summation by parts against the bounded partial sums of x^n proves convergence.

For $|u| = 1$ put

$$\mathcal{B}_j(u) = \mathrm{Li}_j(u) + (-1)^j \mathrm{Li}_j(u^{-1}) \quad (u \neq 1).$$

At $u = 1$ define

$$\mathcal{B}_j(1) = \begin{cases} 2\zeta(j), & j \text{ even}, \\ 0, & j \text{ odd}. \end{cases}$$

In particular $\mathcal{B}_1(1) = 0$ is a symmetric bilateral value; it is not an assignment of a finite value to $\mathrm{Li}_1(1)$. For $u = e^{2\pi i t}$, $0 < t < 1$, the classical inversion identity gives

$$\mathcal{B}_j(u) = -\frac{(2\pi i)^j}{j!} B_j(t),$$

where $B_j(t)$ is the Bernoulli polynomial. Thus all these factors are explicit rational multiples of $(i\pi)^j$ at roots of unity.

For $w = p + q$ define

$$\begin{aligned}\mathcal{R}_{p,q}(x,y) = (-1)^q \Bigg[&\sum_{j=1}^{q} (-1)^j \binom{w-j-1}{p-1} \mathcal{B}_j(r)\,\mathrm{Li}_{w-j}(x) \\ &+ \sum_{j=1}^{p} \binom{w-j-1}{q-1} \mathcal{B}_j(r)\,\mathrm{Li}_{w-j}(y^{-1}) \\ &- \binom{w-1}{p}\mathrm{Li}_w(y^{-1}) - \binom{w-1}{q}\mathrm{Li}_w(x) \\ &- \mathcal{B}_p(x)\,\mathrm{Li}_q(y^{-1}) \Bigg] + (-1)^w\,\mathrm{Li}_w(r^{-1}). \end{aligned} \tag{2.1}$$

No $\mathrm{Li}_1(1)$ occurs in this expression: the arguments of the possible order-one single values are x and y^{-1}, both different from one; the last single value has order $w \geq 2$.

Theorem 2.1 (Explicit bicolored parity, in decreasing-index convention)**.** *For the parameters above,*

$$\boxed{Z_{p,q}(x,y) - (-1)^w Z_{p,q}(x^{-1},y^{-1}) = \mathcal{R}_{p,q}(x,y).} \tag{2.2}$$

Consequently

$$\Re Z_{p,q}(x,y) = \frac{1}{2}\mathcal{R}_{p,q}(x,y) \quad (w \text{ odd}), \qquad \Im Z_{p,q}(x,y) = \frac{1}{2i}\mathcal{R}_{p,q}(x,y) \quad (w \text{ even}).$$

At roots of unity these are finite formulas in single polylogarithms and powers of π. The opposite parity component is not evaluated by this theorem.

2.2 Proof with the summation order made explicit

For $k \geq 1$ form the absolutely convergent bilateral sum

$$C_{p,q}(k;r) = \sum_{m \in \mathbb{Z}\setminus\{0,-k\}} \frac{r^m}{(m+k)^p m^q}.$$

The denominator is $O(|m|^w)$ as $|m| \to \infty$, and $w \geq 2$. The elementary partial-fraction identity

$$\begin{aligned}\frac{1}{(m+k)^p m^q} = &\sum_{j=1}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \frac{1}{k^{w-j} m^j} \\ &+ (-1)^q \sum_{j=1}^{p} \binom{w-j-1}{q-1} \frac{1}{k^{w-j}(m+k)^j} \end{aligned} \tag{2.3}$$

holds away from its poles. It follows either by matching the principal parts at $m = 0, -k$, after which the difference is a rational function with no poles and vanishing at infinity, or by induction on $p + q$.

Sum (2.3) over a symmetric cutoff in m. The possible $j = 1$ terms are legitimate in this order:

$$\lim_{M\to\infty} \sum_{0<|m|\leq M} \frac{r^m}{m^j} = \mathcal{B}_j(r).$$

For $r \neq 1$, the two one-sided $j = 1$ series converge by Dirichlet's test; for $r = 1$ they cancel at every symmetric cutoff. Translating the cutoff by the fixed integer k changes the sum by

finitely many terms tending to zero, so the substitution $m+k=\ell$ is valid. Removing the omitted terms $m=-k$ and $\ell=k$ gives

$$\begin{aligned} C_{p,q}(k;r) = &\sum_{j=1}^{q}(-1)^{q-j}\binom{w-j-1}{p-1}\frac{\mathcal{B}_j(r)}{k^{w-j}} \\ &+(-1)^q r^{-k}\sum_{j=1}^{p}\binom{w-j-1}{q-1}\frac{\mathcal{B}_j(r)}{k^{w-j}} \\ &-\frac{(-1)^q}{k^w}\left[\binom{w-1}{p}r^{-k}+\binom{w-1}{q}\right]. \end{aligned} \tag{2.4}$$

The last line uses the elementary hockey-stick sums $\sum_{j=1}^{q}\binom{w-j-1}{p-1}=\binom{w-1}{p}$ and its p,q counterpart.

For $0<\rho<1$ multiply (2.4) by $\rho^k x^k$ and sum over $k\geq 1$. All the original double sums are now absolutely convergent. On the right, let $\rho\uparrow 1$ in the resulting single polylogarithms; their arguments are x and y^{-1}, neither equal to one. The result is

$$\begin{aligned} \mathcal{A} = (-1)^q\Bigg[&\sum_{j=1}^{q}(-1)^j\binom{w-j-1}{p-1}\mathcal{B}_j(r)\operatorname{Li}_{w-j}(x) \\ &+\sum_{j=1}^{p}\binom{w-j-1}{q-1}\mathcal{B}_j(r)\operatorname{Li}_{w-j}(y^{-1})-\binom{w-1}{p}\operatorname{Li}_w(y^{-1})-\binom{w-1}{q}\operatorname{Li}_w(x)\Bigg]. \end{aligned} \tag{2.5}$$

To identify the left side in the ordinary nested convention, split its lattice points, with $n=m+k$, into three disjoint regions: $n>m>0$, $0>n>m$, and $n>0>m$. They yield, as $\rho\uparrow 1$,

$$\mathcal{A} = Z_{p,q}(x,y)+(-1)^w Z_{q,p}(y^{-1},x^{-1})+(-1)^q\operatorname{Li}_p(x)\operatorname{Li}_q(y^{-1}). \tag{2.6}$$

For completeness, the gap Abel factors ρ^{n-m} used here need not be identified informally with outer-index radial factors. The first region is $Z_{p,q}(\rho x, y/\rho)$ and has the integral

$$Z_{p,q}(a,b) = \frac{1}{(p-1)!}\int_0^1(-\log t)^{p-1}\frac{a\operatorname{Li}_q(abt)}{1-at}\,dt. \tag{2.7}$$

For $a=\rho x$, $ab=r$, its limit follows by dominated convergence: $1-\rho xt$ is bounded away from zero uniformly for ρ near one; $\operatorname{Li}_q(rt)$ is bounded when $q>1$ and at worst logarithmic near $t=1$ when $q=1, r=1$. These are integrable majorants. The same integral identifies this limit with the ordinary nested series via outer-index Abel summation. The negative region is handled identically, and the cross region is the product $(-1)^q\operatorname{Li}_p(\rho x)\operatorname{Li}_q(\rho/y)$. This proves (2.6) for all stated boundary cases.

Finally the convergent stuffle identity gives

$$Z_{q,p}(y^{-1},x^{-1}) = \operatorname{Li}_q(y^{-1})\operatorname{Li}_p(x^{-1})-\operatorname{Li}_w(r^{-1})-Z_{p,q}(x^{-1},y^{-1}).$$

It follows by multiplying finite partial sums and passing to their limits; both nested series have already been shown to converge. Substituting this into (2.6) and using $\mathcal{B}_p(x)=\operatorname{Li}_p(x)+(-1)^p\operatorname{Li}_p(x^{-1})$ gives (2.2). Complex conjugation proves the two displayed real/imaginary consequences.

The collision $r=1$ has therefore been proved directly. One can also see why it is harmless in (2.1): its $\mathcal{B}_1(r)$ coefficient is a constant multiple of $\operatorname{Li}_{w-1}(y^{-1})-\operatorname{Li}_{w-1}(x)$, which vanishes at $r=1$. The jump in the individual bilateral order-one factor cannot create a jump in the complete expression.

2.3 Proof and simplification of the recorded Family–2 identity

Let $L = \log 2$, $G = \beta(2)$, and $v = (1+i)/2$. The exact depth-one values required at weight three are

$$\mathrm{Li}_1(i) = -\frac{L}{2} + \frac{i\pi}{4}, \quad \mathrm{Li}_2(i) = -\frac{\pi^2}{48} + iG, \quad \mathrm{Li}_3(i) = -\frac{3\zeta(3)}{32} + \frac{i\pi^3}{32},$$

with $\mathrm{Li}_n(-i) = \overline{\mathrm{Li}_n(i)}$ and $\mathrm{Li}_n(-1) = -(1-2^{1-n})\zeta(n)$ for $n > 1$. Here $\mathcal{B}_1(-1) = 0$ and $\mathcal{B}_2(-1) = -\pi^2/6$. Inserting these numbers in (2.1), first at $(p,q) = (1,2)$ and then at $(2,1)$, gives

$$\Re\,\mathrm{Li}_{1,2}(i,i) = \frac{33}{64}\zeta(3) - \frac{\pi}{4}G + \frac{\pi^2 L}{24}, \tag{2.8}$$

$$\Re\,\mathrm{Li}_{2,1}(i,i) = \frac{15}{64}\zeta(3) - \frac{\pi^2 L}{32}. \tag{2.9}$$

Consequently the desired difference has the short exact form

$$\boxed{\Re\big(\mathrm{Li}_{1,2}(i,i) - \mathrm{Li}_{2,1}(i,i)\big) = \frac{9}{32}\zeta(3) - \frac{\pi}{4}G + \frac{7}{96}\pi^2 \log 2.} \tag{2.10}$$

To identify this with the exact equation printed in the source, write $\mu_2 = \Re\,\mathrm{Li}_2(v)$, $\lambda_2 = \Im\,\mathrm{Li}_2(v)$ and $h_{a,b} = \Re\,\mathrm{Li}_{a,b}(i,1)$. The classical Landen formula

$$\mathrm{Li}_2\!\left(\frac{z}{z-1}\right) = -\,\mathrm{Li}_2(z) - \frac{1}{2}\log^2(1-z)$$

(applied along the segment $z = it$, $0 \le t \le 1$) gives

$$\mu_2 = \frac{5\pi^2}{96} - \frac{L^2}{8}, \qquad \lambda_2 = G - \frac{\pi L}{8}.$$

Its derivative is an identity of elementary logarithms and its constant is fixed at $z = 0$, so no numerical recognition is needed. The shuffle identity immediately gives

$$h_{1,1} = \frac{L^2}{8} - \frac{\pi^2}{32}.$$

For the remaining height-one value put

$$S(z) = \mathrm{Li}_{2,1}(z,1) = \frac{1}{2}\int_0^z \frac{\log^2(1-t)}{t}\,dt.$$

The needed inversion is already recorded after the proof of the source's Proposition 1.12, "Nielsen inversion with explicit endpoint data" [1]; both the inversion and its Gaussian specialization are also explicit in Panzer's discussion following Corollary 1.4 [5]. To make the present proof independent of an unlabelled source display, let $u = \log(-z)$ on $\mathbb{C} \setminus [0,\infty)$. The identity $\log(1-1/z) = \log(1-z) - u$ shows by differentiation that

$$S(z) + S(1/z) - \mathrm{Li}_3(z) + u\,\mathrm{Li}_2(z) + \frac{u^3}{6}$$

is constant. Here

$$S(1) = \frac{1}{2}\int_0^1 \frac{\log^2 t}{1-t}\,dt = \zeta(3),$$

where the last equality follows by integrating the geometric series. Letting $z \to 1$ along the upper unit semicircle gives $u \to -i\pi$; the two cubic imaginary terms cancel because $\zeta(2) = \pi^2/6$. The constant is therefore $\zeta(3)$, proving

$$S(z) + S(1/z) = \mathrm{Li}_3(z) - \log(-z)\,\mathrm{Li}_2(z) - \frac{1}{6}\log^3(-z) + \zeta(3). \tag{2.11}$$

At $z = i$, conjugation gives $S(1/i) = \overline{S(i)}$ and $\log(-i) = -i\pi/2$, so

$$h_{2,1} = \frac{29}{64}\zeta(3) - \frac{\pi}{4}G.$$

Finally $\mathrm{Li}_1(i)\,\mathrm{Li}_2(i) = \mathrm{Li}_{1,2}(i,1) + 2\,\mathrm{Li}_{2,1}(i,1)$ gives

$$h_{1,2} = \Re\big(\mathrm{Li}_1(i)\,\mathrm{Li}_2(i)\big) - 2h_{2,1} = \frac{\pi^2 L}{96} + \frac{\pi}{4}G - \frac{29}{32}\zeta(3).$$

Direct rational substitution now yields

$$3\zeta(3) - 4L\mu_2 - \pi\lambda_2 - 4Lh_{1,1} + 3h_{1,2} = \frac{9}{32}\zeta(3) - \frac{\pi}{4}G + \frac{7}{96}\pi^2 L.$$

This proves `double:eq:fam2-example` precisely as written, and proves its stronger simplification (2.10). The statement needs no conjectured independence of the three constants.

2.4 Mixed-color examples beyond the displayed source equation

The same finite formula gives the following exact evaluations:

$$\Re\big(\mathrm{Li}_{1,2}(-1,i) - \mathrm{Li}_{2,1}(i,-1)\big) = \frac{15}{16}\zeta(3) - \frac{\pi}{2}G + \frac{\pi^2 L}{48}, \tag{2.12}$$

$$\Im\big(\mathrm{Li}_{1,3}(-1,i) - \mathrm{Li}_{3,1}(i,-1)\big) = \frac{3\pi^3 L}{32} - 3\beta(4) + \frac{21\pi\zeta(3)}{64}, \tag{2.13}$$

$$\Re\big(\mathrm{Li}_{1,4}(-1,i) - \mathrm{Li}_{4,1}(i,-1)\big) = \frac{7\pi^4 L}{11520} - \frac{\pi}{2}\beta(4) + \frac{\pi^2\zeta(3)}{32} + \frac{135}{128}\zeta(5), \tag{2.14}$$

$$\Im\big(\mathrm{Li}_{1,5}(-1,i) - \mathrm{Li}_{5,1}(i,-1)\big) = \frac{5\pi^5 L}{512} - 5\beta(6) + \frac{465\pi\zeta(5)}{1024} + \frac{3\pi^3\zeta(3)}{64}. \tag{2.15}$$

Both indices *and* colors are swapped in these antisymmetric expressions. They are elementary rational specializations of Theorem 2.1, not additional parity conjectures. The accompanying symbolic generator derives the Gaussian formulas at all requested weights; a separate Mellin-integral evaluator checks the left sides without reusing the theorem as its evaluation algorithm.

As a collision check, the case $p = q = 1$, $(x, y) = (i, -i)$ gives

$$\Im\,\mathrm{Li}_{1,1}(i,-i) = G - \frac{\pi\log 2}{8}.$$

Here the symmetric bilateral value $\mathcal{B}_1(1) = 0$ is essential to avoid introducing a spurious divergent $\mathrm{Li}_1(1)$.

2.5 An all-weight family of ordinary logarithmic integrals

The single-inner-index specialization has an additional cancellation: no even beta values remain in its real odd-weight component.

Corollary 2.2 (Odd logarithmic moments)**.** *For $m \geq 1$, put $w = 2m+1$ and*

$$I_m = \frac{1}{(2m-1)!}\int_0^1 \frac{x(-\log x)^{2m-1}\log(1+x)}{1+x^2}\,dx.$$

Then

$$\begin{aligned} I_m = {} & \frac{1}{2}(1-2^{-2m})\left(1-\frac{2m+1}{2^{2m+1}}\right)\zeta(2m+1) \\ & -\sum_{k=1}^{m-1}\frac{(1-2^{1-2k})(1-2^{1-w+2k})}{2^{w-2k}}\zeta(2k)\zeta(w-2k) \\ & -\frac{1}{2}(1-2^{1-2m})(1-2^{-2m})\zeta(2m)\log 2. \end{aligned} \tag{2.16}$$

In particular,

$$\begin{aligned} I_1 &= \frac{15}{64}\zeta(3) - \frac{\pi^2\log 2}{32}, \\ I_2 &= \frac{405}{1024}\zeta(5) - \frac{\pi^2\zeta(3)}{128} - \frac{7\pi^4\log 2}{1536}, \\ I_3 &= \frac{7623}{16384}\zeta(7) - \frac{5\pi^2\zeta(5)}{2048} - \frac{7\pi^4\zeta(3)}{7680} - \frac{31\pi^6\log 2}{61440}. \end{aligned}$$

Proof. Equation (2.7) with $(x,y) = (i,i)$ and $q = 1$ gives $I_m = \Re Z_{2m,1}(i,i)$, since $\mathrm{Li}_1(-t) = -\log(1+t)$ and $\Re(i/(1-it)) = -t/(1+t^2)$. In (2.1) use $\mathcal{B}_{2k}(-1) = 2\,\mathrm{Li}_{2k}(-1)$ and $\mathcal{B}_{2k+1}(-1) = 0$. Writing $u_j = \Re\,\mathrm{Li}_j(i)$, the real part simplifies before any zeta evaluation to

$$2I_m = -2\sum_{k=1}^{m}\mathrm{Li}_{2k}(-1)u_{w-2k} + wu_w - u_{2m}\log 2 - \mathrm{Li}_w(-1).$$

Now $u_1 = -\log 2/2$ and $u_j = -2^{-j}(1-2^{1-j})\zeta(j)$ for $j \geq 2$. Separate the $k = m$ term and substitute these expressions to obtain (2.16). Euler's even-zeta evaluations yield the three displayed cases. □

Two weight-three integral identities also follow directly:

$$\begin{aligned} \int_0^1 \frac{x\log x\log(1+x)}{1+x^2}\,dx &= \frac{\pi^2\log 2}{32} - \frac{15}{64}\zeta(3), \\ \int_0^1 \frac{x\,\mathrm{Li}_2(-x)}{1+x^2}\,dx &= \frac{\pi}{4}G - \frac{\pi^2\log 2}{24} - \frac{33}{64}\zeta(3). \end{aligned}$$

These are ordinary real integrals with their constants fixed by the convergent endpoint representation. They may be useful as simple benchmark identities alongside the bicolored-polylogarithm form.

2.6 Audit conclusion and what this does not settle

The numeric Family–2 example can now be marked proved and replaced, if desired, by (2.10); its older coordinates can be preserved as an exact equivalent identity. No false equality was found in that example. The correction is to its evidence status and to an unnecessarily large choice of constants.

The master identity proves the real components in odd weight and the imaginary components in even weight for every admissible Gaussian mixed pair, and provides the same statement

at all roots of unity. It therefore supplies an exact replacement for searches in these parity components. It does not prove nonreduction in the complementary components. In particular S_6 and S_8 have odd total weights seven and nine and are imaginary components. They lie in the opposite parity sector; this theorem proves neither frozen candidate. The rational weight-nine vector already excluded in the source remains false, and the newer S_6, S_8 candidates remain conjectural. The higher golden ladders are also unaffected.

Attribution safeguard. When comparing formulas, Panzer uses increasing summation indices, so both indices and arguments must be reversed to use his notation. Panzer's footnote 16 explicitly flags a transposed product in Nakamura's printed equation (1.3): the product $\zeta(a;-y)\zeta(b;x)$ must be replaced by $\zeta(a;x)\zeta(b;-y)$ in that convention. The direct proof above rebuilds the products from the three lattice regions and therefore avoids relying on the misprinted equation. This is a literature normalization issue, not a newly discovered source-manuscript error.

3 The optimal moment prefix for adjacent even tail orders

The incoming *Canonical Digamma Polynomials and the Fifth Harmonic Bridge* proves that finitely many initial centered moments generate the full formal coordinate space of two even harmonic tails. For equal orders it proves the optimal consecutive prefix. Its Conjecture `conj:firstq` asks for that optimal prefix for every unequal pair. We prove the entire first off-diagonal family:

$$p = 2b - 2, \qquad q = 2b, \qquad b \geq 2.$$

The proof also gives an exact determinant with a positive rational recurrence, and hence a family of ordinary convergent identities for products of higher polygammas. The result concerns rational coefficient rank. It does not assert arithmetic independence of the constants that those coefficients multiply.

3.1 The inherited exact coordinate system

Write

$$A_p(n) = \zeta(p) - H_n^{(p)} = \zeta(p, n+1), \qquad x_n = n + \tfrac{1}{2},$$

and let D_r be the continued value at $s = -2r$ of

$$D_{p,q}(s) = \sum_{n=0}^{\infty} A_p(n) A_q(n) x_n^{-s}.$$

For even p, q, these particular continued values are regular. The source tail theorem supplies numbers C_j satisfying the inverse Bernoulli transforms

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j, \tag{3.1}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{3.2}$$

For $q = 2b$ and $j \geq b$, they have the finite coordinate expansion

$$C_j = \sum_{k=0}^{b-1} a_{jk} \zeta(2k), \qquad \zeta(0) = -\tfrac{1}{2}, \tag{3.3}$$

where, putting $d = 2j + 1$ and $e = d - p - q + 2k + 1$,

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\} \quad (e \geq 0), \tag{3.4}$$

and $a_{jk} = 0$ if $e < 0$. Each binomial summand in (3.4) is omitted unless its lower index is at least 2 and no greater than its upper index. In particular, a would-be lower index zero is *not* retained. These formulas are prior results of the inspected reports; the new argument is the nonvanishing theorem below.

For reference, the finite mechanism behind the inherited formulas is summation by parts with

$$\sum_{n=0}^{N-1} x_n^{2r} = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) N^{2j+1}, \qquad A_p(k-1) = A_p(k) + k^{-p}.$$

The resulting ordered finite sums reduce by Faulhaber polynomials. Their constant coefficients give (3.3)–(3.4); all subtracted nonnegative even powers have continued sum zero because $\zeta(-2m, 1/2) = 0$. Thus the conventions in (3.1) are tied to the original centered cutoff, rather than to an unspecified regularization.

3.2 A universal positive recurrence

Let

$$h(t) = \coth(t/2) = \sum_{k=0}^{\infty} h_k t^{2k-1}, \qquad h_k = \frac{2B_{2k}}{(2k)!}, \qquad h_0 = 2.$$

The ordinary Bernoulli sign formula gives

$$h_k = (-1)^{k-1} \beta_k, \qquad \beta_k = \frac{2|B_{2k}|}{(2k)!} = \frac{4\zeta(2k)}{(2\pi)^{2k}} > 0 \quad (k \geq 1). \tag{3.5}$$

This is the $x = 0$ case of the classical Bernoulli Fourier formula (DLMF 24.8.1). Define rational numbers c_n by

$$c_1 = 1, \tag{3.6}$$

$$c_n = \frac{1}{4(2n-1)^2} \sum_{r=1}^{n-1} \beta_{n-r} \left\{ 2r(2r-1) + (2n-1)(2n-2) - 2\mathbf{1}_{r=1} \right\} c_r \quad (n \geq 2). \tag{3.7}$$

Every coefficient in this recurrence is strictly positive. Indeed, for $r = 1$ the expression in braces is $(2n-1)(2n-2) > 0$, and for $r \geq 2$ both its displayed summands are positive. Therefore

$$c_n \in \mathbb{Q}_{>0} \quad (n \geq 1). \tag{3.8}$$

The first values are

$$c_1 = 1, \quad c_2 = \frac{1}{36}, \quad c_3 = \frac{11}{5400}, \quad c_4 = \frac{113}{705600}, \quad c_5 = \frac{2509}{190512000}, \quad c_6 = \frac{309713}{276623424000}.$$

Theorem 3.1 (Optimal prefix and exact determinant for adjacent orders)**.** *Let $b \geq 2$, $p = 2b-2$, and $q = 2b$. Define the square late-block matrix*

$$M_b = (a_{b+r,k})_{0 \leq r,k < b}$$

by (3.4). Then

$$\det M_b = (-1)^{b(b-1)/2} \frac{((2b+1)!!)^2}{b!} c_{b+1} \neq 0. \tag{3.9}$$

Consequently $D_0, \dots, D_{q-1}$ have full formal rank q in the coordinate system

$$C_0, \dots, C_{b-1}, \quad \zeta(0), \zeta(2), \dots, \zeta(2b-2).$$

Every later D_r is an explicit rational combination of these first q moments. In this coordinate ordering, the determinant of the complete first-q moment matrix is

$$(-1)^{b(b-1)/2} \frac{((2b+1)!!)^2 c_{b+1}}{b!(4b-1)!!}. \tag{3.10}$$

Proof. For an even polynomial

$$Q(t) = \sum_{n=1}^{b} q_n t^{2n},$$

introduce the operator

$$\mathcal{L}_Q(t) = (Q''(t) - Q''(0))h(t) + (Q(t)h(t))''. \tag{3.11}$$

It is an odd formal power series, and is holomorphic at zero when regarded analytically near zero. The subtraction of $Q''(0)$ is essential: it is precisely the omitted lower-index-zero term in (3.4).

Let K_b be the matrix mapping $(q_1, \dots, q_b)$ to the coefficients of $t, t^3, \dots, t^{2b-1}$ in $\mathcal{L}_Q$. Direct coefficient extraction from (3.4) gives

$$K_b = \operatorname{diag}_{0 \le r < b} \left(\frac{2}{(2r+1)!} \right) M_b^{\mathrm{rev}} \operatorname{diag}_{1 \le n \le b}((2n)!), \tag{3.12}$$

where M_b^{rev} reverses the column order. This can also be read off from the inherited identity $2f^{(2b)} = \mathcal{L}_Q$ for the column generators, but no analytic continuation is needed for the coefficient calculation. More explicitly, with row $r = N-2$ and column n,

$$(K_b)_{N-2,n-1} = h_{N-n}\{2n(2n-1) + (2N-1)(2N-2) - 2\mathbf{1}_{n=1}\}$$

when $n \le N$, and the entry is zero when $n > N$.

Write $Q_*(t) = \sum_{n \ge 1} q_n^* t^{2n}$ for the formal solution of $\mathcal{L}_{Q_*} = 0$ normalized by $q_1^* = 1$. For $n \ge 2$, the coefficient of t^{2n-3} gives

$$4(2n-1)^2 q_n^* + \sum_{r=1}^{n-1} h_{n-r} \{2r(2r-1) + (2n-1)(2n-2) - 2\mathbf{1}_{r=1}\} q_r^* = 0. \tag{3.13}$$

The nonzero coefficient of q_n^* proves existence and uniqueness recursively. Equations (3.5) and (3.7) show that

$$q_n^* = (-1)^{n-1} c_n. \tag{3.14}$$

In particular no coefficient vanishes.

The matrix K_b is upper Hessenberg. Its row corresponding to $n = 2, \dots, b+1$ has the nonzero next-column entry $d_n = 4(2n-1)^2$ whenever that column is present. Successively using these entries to eliminate $q_2, \dots, q_b$ shows

$$\det K_b = (-1)^b q_{b+1}^* \prod_{n=2}^{b+1} d_n = 4^b ((2b+1)!!)^2 c_{b+1}. \tag{3.15}$$

Here the last constraint is the one that would determine q_{b+1}^* in the infinite formal solution; a polynomial has $q_{b+1} = 0$. One explicit determinant argument replaces the first column by

itself plus $q_2^*, \dots, q_b^*$ times the successive columns. Its first $b-1$ entries become zero and its last entry is $-d_{b+1}q_{b+1}^*$. Expanding along that column leaves a lower triangular minor with diagonal $d_2, \dots, d_b$, proving (3.15) with its stated sign. This also provides a determinant-free proof of injectivity: the first $b-1$ constraints force $q_n = q_1 q_n^*$ through $n = b$, and the final one forces $q_1 q_{b+1}^* = 0$, hence $Q = 0$.

Taking determinants in (3.12), and using

$$\prod_{n=1}^{b} \frac{(2n-1)!}{(2n)!} = \frac{1}{2^b b!},$$

proves (3.9). The transform (3.1) is lower triangular with diagonal $1/(2r+1)$, and the first b block rows are the identity on $C_0, \dots, C_{b-1}$. Its product of diagonal entries through $r = 2b-1$ is $1/(4b-1)!!$. This proves (3.10) and the rank assertion. Finally M_b^{-1} reconstructs every even-zeta coordinate from $C_b, \dots, C_{2b-1}$, while (3.2) obtains all these blocks from the first $2b$ moments. Equations (3.4) and (3.1) then reconstruct every later moment by finite rational arithmetic. □

A clarification about the range of the theorem. The determinant formula proves the full infinite family $q - p = 2$. The case $p = q$ was already proved in the incoming report. General unequal gaps $q - p \geq 4$ remain open here; the argument is not a proof of the complete source conjecture. Its positivity mechanism uses a single free initial coefficient, which is specific to the second-order operator in (3.11). Isolated larger-gap pairs, including $(p, q) = (2, 6)$, were already treated in the source; the remaining question is the uniform assertion.

3.3 Ordinary convergent harmonic and polygamma identities

The inverse matrix produces explicit series identities without an unevaluated finite-part symbol. For $0 \leq k < b$, define

$$w_{b,k,r} = \sum_{t=0}^{b-1} (M_b^{-1})_{kt} \binom{2(b+t)+1}{2r} 2^{-2(b+t-r)}, \tag{3.16}$$

$$P_{b,k}(x) = \sum_{r=0}^{2b-1} w_{b,k,r} x^{2r}, \tag{3.17}$$

where a summand in (3.16) is zero if $r > b + t$. Put

$$a_0 = \frac{1}{(2b-3)(2b-1)}, \qquad a_1 = -\frac{2b-2}{24(2b-1)} - \frac{2b}{24(2b-3)}. \tag{3.18}$$

Corollary 3.2 (All even-zeta coordinates from adjacent tails)**.** *For every $b \geq 2$ and $0 \leq k < b$,*

$$\sum_{n=0}^{\infty} \{P_{b,k}(x_n) A_{2b-2}(n) A_{2b}(n) - a_0 w_{b,k,2b-1} x_n^2 - a_0 w_{b,k,2b-2} - a_1 w_{b,k,2b-1}\} = \zeta(2k). \tag{3.19}$$

The series is absolutely convergent. Replacing the tail product by

$$A_{2b-2}(n) A_{2b}(n) = \frac{\psi^{(2b-3)}(n+1)\psi^{(2b-1)}(n+1)}{(2b-3)!(2b-1)!} \tag{3.20}$$

gives the corresponding ordinary polygamma-product identity.

Proof. By (3.3), inversion, and (3.2), $\sum_r w_{b,k,r} D_r = \zeta(2k)$. The centered Hurwitz expansion gives

$$A_{2b-2}(n)A_{2b}(n) = a_0 x_n^{4-4b} + a_1 x_n^{2-4b} + O(x_n^{-4b}).$$

The largest possible degree in (3.17) is $4b-2$. Thus the two displayed subtractions remove every nondecaying term, and the summand is $O(x_n^{-2})$. Centered subtraction computes the same continued moment because the removed monomials have $\zeta(0,1/2) = \zeta(-2,1/2) = 0$. This proves the identity and absolute convergence. Formula (3.20) is the standard Hurwitz-zeta expression for odd-order polygammas. □

3.4 Two exact rational evaluations

For $(p,q) = (4,6)$, the late block matrix and its inverse are

$$M_3 = \begin{pmatrix} 0 & 3/4 & 1/4 \\ 5/12 & 2/3 & -1/12 \\ 1 & -3/8 & 1/12 \end{pmatrix}, \qquad M_3^{-1} = \frac{1}{339}\begin{pmatrix} -28 & 180 & 264 \\ 136 & 288 & -120 \\ 948 & -864 & 360 \end{pmatrix}. \tag{3.21}$$

In particular $\det M_3 = -113/384$. Define

$$\begin{aligned} P_{4,6}(x) = 67 &+ 15516x^2 + 324320x^4 + 1434496x^6 \\ &+ 1601280x^8 + 371712x^{10}. \end{aligned} \tag{3.22}$$

Then the first row of (3.21) after the inverse Bernoulli transform gives the ordinary identity

$$\sum_{n=0}^{\infty} \left\{ P_{4,6}(x_n)A_4(n)A_6(n) - \frac{123904}{5}x_n^2 - \frac{316928}{5} \right\} = -21696. \tag{3.23}$$

The right side is $43392\zeta(0)$. Thus every double-zeta coordinate and every positive even-zeta coordinate cancels exactly.

For $(p,q) = (6,8)$, $\det M_4 = 2509/5120$. Define

$$\begin{aligned} P_{6,8}(x) = 3705 &+ 880180x^2 + 32541904x^4 + 389667904x^6 \\ &+ 1652632320x^8 + 2454887424x^{10} \\ &+ 1189933056x^{12} + 138854400x^{14}. \end{aligned} \tag{3.24}$$

Its rational-coordinate identity is

$$\sum_{n=0}^{\infty} \left\{ P_{6,8}(x_n)A_6(n)A_8(n) - \frac{27770880}{7}x_n^2 - \frac{98910208}{5} \right\} = -7707648. \tag{3.25}$$

The product in (3.23) is $\psi^{(3)}(n+1)\psi^{(5)}(n+1)/720$; the product in (3.25) is $\psi^{(5)}(n+1)\psi^{(7)}(n+1)/604800$. The coefficient files also contain the corresponding exact $\zeta(2), \zeta(4)$, and $\zeta(6)$ rows where applicable.

3.5 A hypergeometric expression for the determinant generator

The formal solution used in the proof has a separate exact expression. It is not needed for nonvanishing, but supplies a second coefficient algorithm. Dividing (3.11) by $2h(t)$ gives

$$Q_*''(t) - \operatorname{csch}(t)Q_*'(t) + \frac{Q_*(t)}{4\sinh^2(t/2)} = 1. \tag{3.26}$$

Set $z = \tanh(t/2)$ and $Q_*(t) = 4z^2F(z^2)$. If $F(u) = \sum_{n\geq 0} f_n u^n$, then

$$f_0 = 1, \qquad (n+\tfrac{1}{2})^2 f_n - n^2 f_{n-1} = \tfrac{1}{4} \quad (n \geq 1), \tag{3.27}$$

and hence

$$f_n = \left(\frac{n!}{(3/2)_n}\right)^2 \sum_{k=0}^{n} \left(\frac{(1/2)_k}{k!}\right)^2. \tag{3.28}$$

This follows by dividing (3.27) by its homogeneous product and telescoping. It is also a direct formal derivation of (3.26); the series converges near the origin, since the coefficients in (3.28) have at most polynomial growth. Consequently

$$\sum_{n=1}^{\infty} (-1)^{n-1} c_n t^{2n} = 4\tanh^2(t/2) \sum_{n=0}^{\infty} f_n \tanh^{2n}(t/2) \tag{3.29}$$

as an analytic identity near zero. The finite binomial-square sum in (3.28) connects the universal determinant coefficients to the classical complete-elliptic-integral hypergeometric series. The present theorem uses this connection as an alternative exact representation, without claiming a new historical priority for that classical series.

3.6 Checks, source audit, and the remaining conjecture

The accompanying `verify_adjacent_tail_prefix.py` compares the original Bernoulli-binomial matrix with an independent differential-operator construction, checks the determinant formula through $b = 16$, and independently reconstructs the two displayed examples from the source U/R block formulas, retaining formal double-zeta coordinates until their exact cancellation. Its numerical option evaluates rational and nonrational rows by direct tail sums followed by an independently generated centered-Hurwitz tail. Those residuals are diagnostics, not interval certificates or substitutes for the proofs above.

No mathematical error in the inherited tail theorem is needed for this advance. Two convention errors are explicitly excluded: omitting $Q''(0)h(t)$ changes the operator, and omitting the column reversal changes the determinant sign. The source optimal-prefix conjecture should be updated to record the newly proved adjacent family, while retaining open status for general gaps at least four.

4 Gamma differential polynomials and exact finite-part reduction

The abstract reduction used in this section is the classical functional-monomial decomposition of differential polynomials. It appears explicitly in Guo–Regensburger–Rosenkranz [6, Proposition 4.10], with an earlier reduction rule in Rosenkranz–Regensburger [7, Lemma 5]. We give its short proof to specify the projection exactly. The special-function contribution is its application to the Gamma hierarchy, the coefficient sectors in the next subsection, and the endpoint and spectral formulas that follow.

4.1 A canonical linear normal form

Let $\mathscr{R} = \mathbb{Q}[u_0, u_1, u_2, \ldots]$, with derivation $Du_r = u_{r+1}$ and weight $\mathrm{wt}(u_r) = r+1$. The factor degree of $\prod u_r^{a_r}$ is $\sum a_r$. A monomial is called functional if it is a power of u_0 (including 1), or its largest positive derivative index occurs at least twice. Let $\mathscr{F}$ be their rational span.

Theorem 4.1 (Classical normal form, with a fixed primitive)**.** *Every $P \in \mathscr{R}$ has a unique decomposition*

$$P = \mathcal{N}(P) + D\mathcal{H}(P), \qquad \mathcal{N}(P) \in \mathscr{F}, \qquad \mathcal{H}(P) \text{ has zero constant term.} \tag{4.1}$$

Both operators are rational and linear. Reduction preserves factor degree; $\mathcal{N}$ preserves weight and $\mathcal{H}$ lowers weight by one. If $p(m)$ is the partition function, then

$$\dim_{\mathbb{Q}} \mathscr{F}_m = p(m) - p(m-1) \quad (m \geq 2), \qquad \dim \mathscr{F}_0 = \dim \mathscr{F}_1 = 1. \tag{4.2}$$

Proof. If a monomial M has a unique largest index $r \geq 1$, replace that factor u_r by u_{r-1}, obtaining $\widetilde{M}$. If a is the multiplicity of u_{r-1} in $\widetilde{M}$, then

$$M = \frac{1}{a} D\widetilde{M} + \left(M - \frac{1}{a} D\widetilde{M} \right).$$

Every remaining monomial has largest derivative index at most $r-1$: all differentiations of a factor u_{r-1} give precisely the a copies of M that were removed. Induction proves existence and termination. It also proves the grading assertions.

For uniqueness, a nonconstant polynomial H of largest derivative index r has $DH = u_{r+1}\partial H/\partial u_r +$ terms of order at most r. Its terms of order $r+1$ have that largest index exactly once and cannot belong to $\mathscr{F}$. Hence $D\mathscr{R} \cap \mathscr{F} = 0$, and the kernel of D consists of constants. Fixing the constant of H proves uniqueness. There are $p(m)$ monomials of weight m; for $m \geq 2$, D is injective on the positive-weight space of weight $m-1$. This proves (4.2). □

The quotient $\mathscr{R}/D\mathscr{R}$ in this theorem is a *vector-space quotient*. For example $\mathcal{N}(u_1) = 0$, but $\mathcal{N}(u_1^2) = u_1^2$. Thus $\mathcal{N}$ is not a ring homomorphism. No multiplication of congruences modulo $D\mathscr{R}$ is permitted without a separate justification.

Now put $u_r = \psi^{(r)}(x)$. At zero the convergent Laurent expansion is

$$\psi(x) = -\frac{1}{x} - \gamma + \sum_{j \geq 1} (-1)^{j+1} \zeta(j+1) x^j, \qquad |x| < 1. \tag{4.3}$$

It follows from the standard log-Gamma series [9, §5.7]; all higher local expansions follow by differentiation. Define the explicit endpoint functional

$$\Delta H = H(\psi(1), \psi'(1), \ldots) - [x^0] H(\psi(x), \psi'(x), \ldots). \tag{4.4}$$

Then (1.7) gives

$$\mathrm{FP} \int_0^1 P(\psi, \psi', \ldots)\, \mathrm{d}x = \mathrm{FP} \int_0^1 \mathcal{N}(P)(\psi, \psi', \ldots)\, \mathrm{d}x + \Delta\mathcal{H}(P). \tag{4.5}$$

This is a finite algebraic recipe: for a monomial of weight w, write it as $x^{-w} \prod_r (x^{r+1}\psi^{(r)}(x))^{a_r}$ and take its coefficient of x^w. Only terms through degree w in these ordinary series are needed. The resulting endpoint belongs to $\mathbb{Q}[\gamma, \zeta(2), \zeta(3), \ldots]$.

4.2 The inverse-Gamma hierarchy and its balance identity

Define

$$\mathcal{A}_m(x) = (-1)^m \Gamma(x) \frac{d^m}{dx^m} \frac{1}{\Gamma(x)}, \qquad \mathcal{A}_0 = 1. \tag{4.6}$$

As universal differential polynomials,

$$\mathcal{A}_{m+1} = u_0\mathcal{A}_m - D\mathcal{A}_m, \qquad \sum_{m\geq 0}\mathcal{A}_m\frac{t^m}{m!} = \exp\left(\sum_{r\geq 0}\frac{(-1)^r u_r t^{r+1}}{(r+1)!}\right). \tag{4.7}$$

The generating function is simply $\Gamma(x)/\Gamma(x-t)$. Writing $B_j = \Gamma^{(j)}(x)/\Gamma(x)$ gives the exact balance identity

$$\mathcal{A}_m = B_r\mathcal{A}_{m-r} - D\sum_{j=0}^{r-1} B_j\mathcal{A}_{m-1-j}, \qquad 0\leq r\leq m. \tag{4.8}$$

Indeed $DB_j = B_{j+1} - u_0B_j$ and $D\mathcal{A}_k = u_0\mathcal{A}_k - \mathcal{A}_{k+1}$, so differentiating the sum telescopes. Also $B_m(u) = (-1)^m\mathcal{A}_m(-u)$. Taking $r = m$ in (4.8) and then the normal form proves

$$\mathcal{N}(\mathcal{A}_m)(u) = (-1)^m\mathcal{N}(\mathcal{A}_m)(-u). \tag{4.9}$$

Only factor degrees congruent to m modulo two survive.

The first reductions illustrate what must still be integrated:

$$\begin{aligned}
&\mathcal{N}(\mathcal{A}_2) = u_0^2, && \mathcal{N}(\mathcal{A}_3) = u_0^3,\\
&\mathcal{N}(\mathcal{A}_4) = u_0^4 - u_1^2, && \mathcal{N}(\mathcal{A}_5) = u_0^5 - 5u_0u_1^2,\\
&\mathcal{N}(\mathcal{A}_6) = u_0^6 - 15u_0^2u_1^2 + u_2^2, &&\\
&\mathcal{N}(\mathcal{A}_7) = u_0^7 - 35u_0^3u_1^2 + 7u_0u_2^2, &&\\
&\mathcal{N}(\mathcal{A}_8) = u_0^8 - 70u_0^4u_1^2 + 28u_0^2u_2^2 - 7u_1^4 - u_3^2. &&
\end{aligned} \tag{4.10}$$

The full differential identities, including primitives, are exact polynomial identities. The accompanying code supplies them through weight ten; the same finite algorithm applies at every weight.

4.3 The inherited scalar bridge, with its normalization proved

Let

$$\frac{1}{\Gamma(1+u)} = \sum_{j\geq 0} c_j u^j, \qquad c_0 = 1, \quad c_1 = \gamma. \tag{4.11}$$

The source reports establish the all-depth Gamma-quotient bridge [2, §3]. We reproduce a proof here because its Laurent sign and scalar normalization enter the sixth moment. Define

$$\mathcal{C}_m(x) = m![t^m]\frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)}. \tag{4.12}$$

For $m\geq 2$ it extends continuously to $[0,1]$, and

$$\int_0^1 \mathcal{C}_m(x)\,\mathrm{d}x = m!(c_m - \lambda_{m-1}). \tag{4.13}$$

Here is a direct Mellin normalization. Set $L(v) = -\log(1-e^{-v})$ and

$$V_d(v) = L(v)^d - \mathbf{1}_{(0,1)}(v)(-\log v)^d.$$

The generating identity for strict unit-index harmonic sums and integration by parts give, initially for $\operatorname{Re} u > 0$,

$$\zeta(1+u,\{1\}^{d-1}) = \frac{1}{\Gamma(1+u)u^d} + \frac{u}{d!\Gamma(1+u)}\int_0^\infty v^{u-1}V_d(v)\,\mathrm{d}v. \tag{4.14}$$

For clarity, the unsplit integral is $[\Gamma(u)d!]^{-1}\int_0^\infty v^{u-1}L(v)^d\,\mathrm{d}v$. It follows by using

$$\sum_{n\ge1} H_{n-1}(\{1\}^{d-1})e^{-nv} = \frac{e^{-v}L(v)^{d-1}}{(d-1)!(1-e^{-v})} = -\frac{1}{d!}\frac{d}{dv}L(v)^d.$$

The subtracted integral in (4.14) is holomorphic near $u=0$: near zero $V_d(v) = O(v(1+|\log v|)^{d-1})$, and at infinity it decays exponentially. Consequently

$$\lambda_d = c_{d+1} + \frac{1}{d!}\int_0^\infty \frac{V_d(v)}{v}\,\mathrm{d}v. \tag{4.15}$$

For the other side put $F(x,t) = \Gamma(1+t)\Gamma(x)/\Gamma(x+t)$ and

$$\Phi(t) = \int_0^1 \{F(x,t) - 1 + t(\psi(x)+\gamma)\}\,\mathrm{d}x.$$

The residue t/x of F cancels the residue $-t/x$ of the last term. The remaining integrand is jointly holomorphic in t near zero and continuous in $x\in[0,1]$. Euler's beta integral, combined with the digamma integral before interchanging integrations, yields

$$\Phi(t) = t\int_0^\infty \frac{(1-e^{-v})^t - 1 - \mathbf{1}_{(0,1)}(v)(v^t-1)}{v}\,\mathrm{d}v. \tag{4.16}$$

To verify the subtraction constant, for $\operatorname{Re} t > 0$ integration in x first gives

$$t\int_0^\infty \left\{\frac{(1-e^{-v})^t - 1}{v} + \frac{e^{-v}}{1-e^{-v}}\right\}\mathrm{d}v - 1.$$

With lower cutoff δ, the second integral is $-\log(1-e^{-\delta})$ and $\int_\delta^1 (v^t-1)\,\mathrm{d}v/v = (1-\delta^t)/t + \log\delta$. Their limits give exactly (4.16). The latter integral continues holomorphically to a neighborhood of zero by its displayed local subtraction. Comparing the coefficient of t^m on its two sides gives $\int \mathcal{C}_m = -m\int V_{m-1}(v)\,\mathrm{d}v/v$, proving (4.13). This proof also justifies all coefficient extractions as ordinary integrals.

Theorem 4.2 (All inverse-Gamma finite parts)**.** *For $m\ge2$ put $\mathcal{L}_m = \mathrm{FP}\int_0^1 \mathcal{A}_m(x)\,\mathrm{d}x$. Then*

$$\boxed{\mathcal{L}_m = K_m - m!\sum_{d=1}^{m-1}(-1)^{m-1-d}c_{m-1-d}\lambda_d, \qquad K_{2n} = \frac{(-1)^n\pi^{2n}}{2n+1}, \quad K_{2n+1}=0.} \tag{4.17}$$

This is the scalar bridge already implicit in the incoming all-order Gamma formulas; the normal-form and spectral applications below are the extensions developed here.

Proof. The coefficientwise finite part of the normalized quotient is

$$1 + \gamma t + \sum_{m\ge2}(c_m - \lambda_{m-1})t^m = \frac{1}{\Gamma(1+t)} - \sum_{d\ge1}\lambda_d t^{d+1}.$$

Here $\mathrm{FP}\int \mathcal{C}_1 = \gamma$, since $\mathcal{C}_1 = \psi+\gamma$ and $\mathrm{FP}\int\psi = 0$. Multiply by $1/\Gamma(1-t) = \sum(-1)^j c_j t^j$. Gamma reflection turns the scalar product into $\sin(\pi t)/(\pi t)$. Extracting $m![t^m]$ proves (4.17). □

If $\mathcal{N}(\mathcal{A}_m) = u_0^m + R_m$ and $H_m = \mathcal{H}(\mathcal{A}_m)$, the immediate all-order moment identity is

$$\boxed{Q_m := \mathrm{FP}\int_0^1 \psi(x)^m\,\mathrm{d}x = \mathcal{L}_m - \Delta H_m - \mathrm{FP}\int_0^1 R_m(\psi,\psi',\ldots)\,\mathrm{d}x.} \tag{4.18}$$

Every monomial of R_m contains a derivative. Section 6 will evaluate that remainder by convergent harmonic coordinates and provide its explicit antiderivative, so (4.18) is part of a terminating exact procedure rather than an instruction to leave mixed integrals unnamed.

4.4 Two exact coefficient sectors at every degree

Let $\mathcal{A}_m$ be defined by

$$\mathcal{A}_0 = 1, \qquad \mathcal{A}_{m+1} = u_0\mathcal{A}_m - D\mathcal{A}_m, \qquad Du_r = u_{r+1}.$$

Equivalently its exponential generating series is

$$\mathcal{A}(t) = \sum_{m\geq 0} \mathcal{A}_m \frac{t^m}{m!} = \exp\left(\sum_{r\geq 0} \frac{(-1)^r u_r t^{r+1}}{(r+1)!}\right). \tag{4.19}$$

All series in this subsection are formal power series. Therefore each coefficient computation is finite. Write $\mathcal{N}$ for the classical functional-monomial projection along the vector space $D\mathbb{Q}[u_0, u_1, \ldots]$. It retains powers of u_0 and monomials in which the largest positive derivative index occurs at least twice. This projection is linear and need not respect products.

Lemma 4.3 (Translation and centered generation)**.** *For an independent constant c,*

$$\mathcal{N}(\mathcal{A}_m)(u_0 + c, u_1, \ldots) = \sum_{j=0}^{m} \binom{m}{j} c^{m-j} \mathcal{N}(\mathcal{A}_j)(u_0, u_1, \ldots). \tag{4.20}$$

Also

$$\mathcal{N}\mathcal{A}(t) = \mathcal{N}\exp\left(\sum_{k\geq 0} \frac{u_{2k} t^{2k+1}}{2^{2k}(2k+1)!}\right). \tag{4.21}$$

Proof. Substitution $u_0 \mapsto u_0 + c$ commutes with D, and preserves the span of the functional monomials. Uniqueness of the direct-sum decomposition therefore implies that it commutes with $\mathcal{N}$. Formula (4.19) acquires the factor e^{ct} under this substitution, proving (4.20).

For the second assertion, apply the formal translation $e^{tD/2}$ to (4.19). Directly summing its exponent gives

$$e^{tD/2}\mathcal{A}(t) = \exp\left(\sum_{k\geq 0} \frac{u_{2k} t^{2k+1}}{2^{2k}(2k+1)!}\right).$$

Every coefficient of $(e^{tD/2} - 1)\mathcal{A}(t)$ belongs to the image of D, proving the claim. In particular every surviving monomial has factor degree congruent to its weight modulo two: the centered exponent contains only variables of odd weight. □

Theorem 4.4 (All polygamma-square coefficients)**.** *For every $r \geq 1$ and $m \geq 2r + 2$,*

$$[u_0^{m-2r-2} u_r^2]\,\mathcal{N}(\mathcal{A}_m) = (-1)^r \binom{m}{2r+2}. \tag{4.22}$$

Thus every square term with exactly two factors of positive derivative order is known at all weights.

Proof. The operator D and the reduction preserve factor degree. At weight $2r + 2$, the quadratic part of $\mathcal{A}_{2r+2}$ is

$$\frac{(2r+2)!}{2} \sum_{a+b=2r} \frac{u_a u_b}{(a+1)!(b+1)!}.$$

Repeated integration by parts gives $\mathcal{N}(u_a u_{2r-a}) = (-1)^{a-r} u_r^2$. The coefficient in question is consequently

$$\frac{1}{2}\sum_{j=1}^{2r+1}(-1)^{j-1-r}\binom{2r+2}{j} = (-1)^r,$$

because the full alternating binomial sum is zero and its two endpoint terms are both one. Formula (4.20), with the original u_0 set to zero, now supplies the factor $\binom{m}{2r+2}$. □

Theorem 4.5 (The complete digamma–trigamma sector)**.** *After the normal form has been computed, set $u_r = 0$ for every $r \geq 2$. Then, for all $m \geq 0$,*

$$\mathcal{N}(\mathcal{A}_m)|_{u_2=u_3=\cdots=0} = u_0^m - \sum_{s=1}^{\lfloor m/4 \rfloor}\binom{m}{4s}\frac{(4s-1)!!}{4s^2-1}u_0^{m-4s}u_1^{2s}. \tag{4.23}$$

Equivalently, put $y = u_1 t^2/2$ and define the entire function $\mathrm{Shi}(y) = \int_0^y \sinh(v)\,dv/v$. The exponential generating series of (4.23) is

$$\sum_{m\geq 0}\mathcal{N}(\mathcal{A}_m)\frac{t^m}{m!}\bigg|_{u_2=u_3=\cdots=0} = \frac{e^{u_0 t}}{2}\left(\cosh y + \frac{\sinh y}{y} - y\,\mathrm{Shi}(y)\right), \tag{4.24}$$

where $\sinh y/y$ has its removable value one at zero.

Proof. Let I be the monomial ideal generated by $u_r u_s$ for $r, s \geq 2$. It is stable under D. The stronger property needed here is

$$\mathcal{N}(I) \subseteq I. \tag{4.25}$$

Indeed, an irreducible monomial in I is retained. If a reducible monomial in I has a unique largest index r, then $r \geq 3$. Replacing u_r by u_{r-1} in its reduction primitive preserves at least two factors of index at least two. The primitive and its derivative therefore remain in I. Induction through the reduction proves (4.25). This argument is essential; stability under D alone would not imply stability under an arbitrary projection.

Modulo I, the Bell series contains at most one factor of derivative order at least two:

$$\mathcal{A}(t) \equiv e^{u_0 t - u_1 t^2/2}\left(1 + \sum_{r\geq 2}\frac{(-1)^r u_r t^{r+1}}{(r+1)!}\right) \pmod{I}. \tag{4.26}$$

For integers $l, k \geq 0$ and $r \geq 2$, integration by parts yields, modulo the vector subspace $D\mathbb{Q}[u_0, u_1, \ldots] + I$,

$$u_0^l u_1^k u_r \equiv \frac{(-1)^{r-1}(l)_{\underline{r-1}}}{k+r-1}u_0^{l-r+1}u_1^{k+r} \quad (l \geq r-1). \tag{4.27}$$

For $l < r-1$ the class is zero. Here the underline denotes a falling factorial. To prove the formula, for $r \geq 3$ differentiate $u_0^l u_1^k u_{r-1}$: the term containing $u_2 u_{r-1}$ belongs to I, leaving a factor $-l$ and reducing r by one. The final step is

$$u_0^l u_1^k u_2 \equiv -\frac{l}{k+1}u_0^{l-1}u_1^{k+2}.$$

If there are too few factors of u_0, a zero coefficient appears before this last step.

Let $d_M = [u_1^M]\mathcal{N}(\mathcal{A}_{2M})$ for $M \geq 2$. The term without a higher derivative in (4.26) contributes $(2M)!(-1)^M/(2^M M!)$. For a term with one u_r, the necessary indices are $l = r-1$ and $k = M-r$, where $2 \leq r \leq M$. Formula (4.27) therefore gives

$$\begin{aligned}\frac{d_M}{(2M)!} &= \frac{(-1)^M}{2^M M!} + \frac{(-1)^M}{M-1}\sum_{r=2}^{M}\frac{(-1)^{r-1}}{(M-r)!2^{M-r}(r+1)!} \\ &= \frac{(-1)^M((-1)^{M+1}-1)}{2^{M+1}(M-1)(M+1)!}.\end{aligned}$$

For the last equality, use the binomial expansion of $(1/2-1)^{M+1}$ and remove its terms of indices zero, one and two. Thus $d_M = 0$ for odd M, while

$$d_M = -\frac{(2M-1)!!}{M^2-1} \qquad (M \geq 2 \text{ even}).$$

The pure u_0 coefficient is one; there is no retained linear u_1 term. The Appell identity now proves (4.23).

Finally, its generating factor is

$$1 - \sum_{s\geq 1}\frac{y^{2s}}{(2s)!((2s)^2-1)}.$$

Expanding $\cosh y$, $\sinh y/y$ and $y\,\mathrm{Shi}(y)$ shows that this is the bracket in (4.24) divided by two. This also proves that the displayed apparent singularity is removable. □

The order of operations in (4.23) is part of its statement: normal reduction precedes specialization. No assertion is being made that the normal projection is multiplicative or that one may impose $u_2 = u_3 = \cdots = 0$ before integration by parts. On substituting $u_r = \psi^{(r)}(x)$, these formulas give every digamma–trigamma coefficient and every pure polygamma-square coefficient in the finite-part moment reductions. All omitted normal-form terms in (4.23) contain at least two factors of derivative order at least two. Their integrals and the exact endpoint functional must still be retained in the full identity.

5 Hurwitz products, spectral jets, and Stieltjes derivative correlations

For integers $k, l \geq 2$, put $W = k + l$ and $C_{kl} = \binom{W-2}{k-1}$. In the original coordinate x, define

$$I_{kl} = \mathrm{FP}_{\epsilon\downarrow 0}\int_{\epsilon}^{1}\zeta(k,x)\zeta(l,x)\,dx.$$

Here the finite part retains the constant coefficient after all negative powers of ϵ and all logarithmic divergences are removed. Then

$$I_{kl} = \begin{cases}(-1)^k C_{kl}\{[2H_{W-2} - H_{k-1} - H_{l-1}]\zeta(W-1) + 2\zeta'(W-1)\}, & k \equiv l \pmod 2,\\ (-1)^k C_{kl}[H_{l-1} - H_{k-1}]\zeta(W-1), & k \not\equiv l \pmod 2.\end{cases}$$

In particular

$$I_{22} = 2\zeta(3) + 4\zeta'(3), \qquad I_{33} = -7\zeta(5) - 12\zeta'(5), \qquad I_{23} = \tfrac{3}{2}\zeta(4), \qquad I_{24} = \tfrac{16}{3}\zeta(5) + 8\zeta'(5).$$

For polygamma products one multiplies I_{kl} by $(-1)^{k+l}(k-1)!(l-1)!$.

Proof and normalization. The classical product kernel [8, Remark 1], also recorded in [1, Chapter 7], underlies the following proof. The classical Mikolás/Hurwitz product integral, already reproduced in the ProveIt manuscript's coincident-endpoint chapter, continues meromorphically as

$$\begin{aligned} F(s,t) &= 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2}\cos\left(\frac{\pi(s-t)}{2}\right)\zeta(2-s-t) \\ &= \frac{\pi\Gamma(s+t-1)\zeta(s+t-1)}{\Gamma(s)\Gamma(t)}\{\csc(\pi s)+\csc(\pi t)\}. \end{aligned}$$

The second expression follows from the Riemann functional equation and Gamma reflection. To relate its Laurent coefficient to the coordinate finite part, write

$$\zeta(s,x) = x^{-s} + \sum_{j=0}^{J} c_j(s)x^j + O(x^{J+1}), \qquad c_j(s) = \frac{(-1)^j(s)_j}{j!}\zeta(s+j),$$

uniformly for s in a small neighborhood of k, with J sufficiently large. There is an analogous expansion near l. Uniformly subtract the resulting local terms before integrating. Along $s = k+u, t = l+u$, the only local terms whose exponent crosses -1 are

$$c_{k-1}(l+u)x^{-1-u}, \qquad c_{l-1}(k+u)x^{-1-u}.$$

Their analytic integrals are respectively $-c_{k-1}(l+u)/u$ and $-c_{l-1}(k+u)/u$, whereas their coordinate finite parts at $u=0$ are zero. All other subtracted monomials have denominators nonzero at zero and commute with evaluation. Therefore

$$I_{kl} = [u^0]F(k+u,l+u) + c'_{k-1}(l) + c'_{l-1}(k).$$

This is the precise local correction; extracting a spectral Laurent constant alone would give the wrong coordinate finite part.

If k, l have the same parity, let $\sigma = (-1)^k = (-1)^l$. Expansion of the cosecant representation gives

$$[u^0]F(k+u,l+u) = 2\sigma C_{kl}\{[2\psi(W-1)-\psi(k)-\psi(l)]\zeta(W-1)+2\zeta'(W-1)\}.$$

The two coefficient derivatives have sum equal to the negative of one half of this expression. If k, l have opposite parity, the two cosecants cancel identically on the chosen line, so its Laurent constant is zero; the coefficient derivatives give $(-1)^k C_{kl}[\psi(l)-\psi(k)]\zeta(W-1)$. Finally $\psi(n) = H_{n-1} - \gamma$ proves the displayed formulas.

5.1 All order derivatives from a holomorphic local generator

The coordinate correction extends to every independent order derivative. Continue to write $c_j(s) = (-1)^j(s)_j\zeta(s+j)/j!$, and define

$$\mathscr{H}_{kl}(s,t) = F(s,t) + \frac{c_{k-1}(t)}{s-k} + \frac{c_{l-1}(s)}{t-l}. \tag{5.1}$$

The singularities displayed separately in this expression are removable in their sum near (k,l).

Theorem 5.1 (Every independent order jet)**.** *For every $k, l \geq 2$, $\mathscr{H}_{kl}$ is holomorphic in a neighborhood of (k,l). For all nonnegative integers a, b,*

$$\partial_s^a\partial_t^b\mathscr{H}_{kl}(s,t)\Big|_{s=k,t=l} = \mathrm{FP}\int_0^1 \partial_s^a\zeta(s,x)|_{s=k}\,\partial_t^b\zeta(t,x)\Big|_{t=l}\,dx. \tag{5.2}$$

The finite part is taken in the same original coordinate x for all orders. Every jet on the left can be obtained by finite Taylor arithmetic in Gamma functions and the ordinary Riemann zeta function.

Proof. Choose small neighborhoods of k and l avoiding the spectral pole at 1, and subtract a sufficiently long Taylor polynomial of each $g_s(x) = \zeta(s, 1+x)$ at $x = 0$. The product consists of an integrable holomorphic remainder, the monomial x^{-s-t}, and finitely many cross monomials $c_j(t)x^{j-s}$ and $c_j(s)x^{j-t}$. Near (k, l), all denominators obtained by integrating these monomials are nonzero except $k - s$ in the first cross family and $l - t$ in the second. Those two integrals are exactly $-c_{k-1}(t)/(s - k)$ and $-c_{l-1}(s)/(t - l)$; adding them back proves holomorphy of (5.1).

Uniform remainder estimates permit every fixed number of parameter derivatives under the integrable remainder: the resulting bounds have the form $Cx^{-\delta}(1 + |\log x|^N)$ with $0 < \delta < 1$. For each nonresonant monomial the coordinate finite part and parameter differentiation agree, since its denominator is nonzero. All derivatives of either removed resonant monomial, evaluated at (k, l), are constant multiples of $x^{-1}\log^j x$. Their integrals over $[\epsilon, 1]$ have constant coefficient zero for every $j \geq 0$. Thus they contribute zero to the fixed-coordinate finite part at every jet order, proving (5.2). □

There is a useful version of (5.1) with its removable singularities already resolved. Put

$$P(s,t) = \frac{\Gamma(s+t-1)}{\Gamma(s)\Gamma(t)}\zeta(s+t-1), \quad E(u) = \frac{\pi u}{\sin(\pi u)}, \quad \sigma_k = (-1)^k,$$

and write $s = k + u$, $t = l + v$. Since $c_{k-1}(t) = -\sigma_k P(k, t)$, one obtains

$$\begin{aligned}\mathscr{H}_{kl}(k+u, l+v) = \sigma_k &\frac{P(k+u,l+v)E(u) - P(k,l+v)}{u} \\ &+ \sigma_l \frac{P(k+u,l+v)E(v) - P(k+u,l)}{v}.\end{aligned} \tag{5.3}$$

Both displayed quotients are holomorphic divided differences. If

$$P(k+u, l+v) = \sum_{a,b\geq 0} P_{ab}u^a v^b, \qquad E(u) = \sum_{r\geq 0} e_{2r}u^{2r},$$

then $e_0 = 1$, $e_2 = \pi^2/6$, $e_4 = 7\pi^4/360$, and the jet in (5.2) is the finite expression

$$a!b!\left\{\sigma_k \sum_{r=0}^{\lfloor (a+1)/2\rfloor} e_{2r}P_{a+1-2r,b} + \sigma_l \sum_{r=0}^{\lfloor (b+1)/2\rfloor} e_{2r}P_{a,b+1-2r}\right\}. \tag{5.4}$$

Only Gamma/zeta Taylor coefficients through total degree $a + b + 1$ are needed. This gives an explicit terminating procedure rather than an instruction to evaluate a difference of poles numerically.

5.2 A first derivative identity with every correction

Let

$$A_k = H_{2k-2} - H_{k-1}, \qquad H_n^{(2)} = \sum_{j=1}^{n} \frac{1}{j^2}.$$

The first diagonal order jet is

$$\begin{aligned}&\mathrm{FP}\int_0^1 \zeta(k,x)\,\partial_s\zeta(s,x)|_{s=k}\,dx \\ &= \frac{(-1)^k}{2}\binom{2k-2}{k-1}\Big\{\left[3A_k^2 + 4\zeta(2) - 3H_{2k-2}^{(2)} + H_{k-1}^{(2)}\right]\zeta(2k-1) \\ &\qquad + 6A_k\zeta'(2k-1) + 3\zeta''(2k-1)\Big\}.\end{aligned} \tag{5.5}$$

In particular,

$$\mathrm{FP}\int_0^1 \zeta(2,x)\,\partial_s\zeta(s,x)|_{s=2}\,dx = (4\zeta(2)-2)\zeta(3)+3\zeta'(3)+3\zeta''(3). \tag{5.6}$$

Proof. Set

$$B(s)=\frac{\Gamma(2s-1)}{\Gamma(s)^2}\zeta(2s-1), \qquad D(s)=\frac{\Gamma(s+k-1)}{\Gamma(s)\Gamma(k)}\zeta(s+k-1).$$

Along the diagonal through (k,k), the functional equation and (5.1) give

$$\mathscr{H}_{kk}(s,s)=2\pi\csc(\pi s)B(s)-\frac{2(-1)^k D(s)}{s-k}.$$

Since $B(k)=D(k)$, the coefficient of $s-k$ is

$$(-1)^k\left\{B''(k)-D''(k)+\frac{\pi^2}{3}B(k)\right\}.$$

Symmetry of $\mathscr{H}_{kk}$ makes the desired single order derivative one half of that coefficient. Logarithmic differentiation of the two Gamma quotients yields

$$\begin{aligned}\frac{B''(k)}{C_{kk}} &= \left[4A_k^2+2\zeta(2)-4H^{(2)}_{2k-2}+2H^{(2)}_{k-1}\right]\zeta(2k-1)\\ &\quad +8A_k\zeta'(2k-1)+4\zeta''(2k-1),\\ \frac{D''(k)}{C_{kk}} &= \left[A_k^2-H^{(2)}_{2k-2}+H^{(2)}_{k-1}\right]\zeta(2k-1)+2A_k\zeta'(2k-1)+\zeta''(2k-1).\end{aligned}$$

Subtraction and $\pi^2/3=2\zeta(2)$ prove (5.5); the $k=2$ specialization gives (5.6). □

5.3 Products of ordinary derivatives of generalized Stieltjes constants

The completed kernel also evaluates all quadratic products of positive ordinary derivative orders of generalized Stieltjes constants. Use the standard convention

$$\zeta(1+u,x)=\frac{1}{u}+\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\gamma_n(x)u^n.$$

For integers $r\ge 1$ and $0\le j\le r$, put

$$p_{r,j}=\left.\frac{d^j}{du^j}(1+u)_r\right|_{u=0},$$

and define the constant-coefficient differential operator

$$T^{(s)}_{n,r}=\sum_{j=0}^{\min(n,r)}\binom{n}{j}p_{r,j}\partial_s^{n-j}. \tag{5.7}$$

Its coefficients are explicit positive integers; in particular $p_{r,0}=r!$. They may equally be computed from $p_{r,j}=j!\,[u^j](1+u)_r$.

Corollary 5.2 (Every quadratic derivative-Stieltjes finite part)**.** *For all Stieltjes indices $m, n \geq 0$ and positive ordinary derivative orders $r, t \geq 1$,*

$$\begin{aligned} &\mathrm{FP}\int_0^1 \gamma_m^{(r)}(x)\gamma_n^{(t)}(x)\,dx \\ &\quad = (-1)^{m+n+r+t}\, T_{m,r}^{(s)} T_{n,t}^{(v)}\, \mathscr{H}_{r+1,t+1}(s,v)\Big|_{s=r+1,v=t+1}. \end{aligned} \tag{5.8}$$

The right side is evaluated by the finite algorithm (5.4). Superscripts (r) and (t) on γ mean ordinary derivatives with respect to its argument x, not derivatives with respect to its Stieltjes index.

Proof. For $r \geq 1$, differentiation with respect to x eliminates the spectral pole $1/u$. The Hurwitz differentiation formula gives

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\gamma_n^{(r)}(x)u^n = (-1)^r(1+u)_r\zeta(r+1+u,x).$$

Taking the coefficient of u^n and applying the product rule yields

$$\gamma_n^{(r)}(x) = (-1)^{n+r}\, T_{n,r}^{(s)}\zeta(s,x)\Big|_{s=r+1}. \tag{5.9}$$

Multiply the two finite expressions (5.9), use linearity of the specified coordinate finite part, and apply Theorem 5.1 term by term. This proves (5.8). □

For the first nontrivial mixed-index example,

$$\gamma_0'(x) = -\zeta(2,x), \qquad \gamma_1'(x) = \zeta(2,x) + \partial_s\zeta(s,x)|_{s=2}.$$

Combining I_{22} with (5.6) gives

$$\boxed{\mathrm{FP}\int_0^1 \gamma_0'(x)\gamma_1'(x)\,dx = -4\zeta(2)\zeta(3) - 7\zeta'(3) - 3\zeta''(3).} \tag{5.10}$$

This family is a consequence of the completed classical product kernel and the standard Hurwitz derivative formula. Its range here is explicitly $r, t \geq 1$; undifferentiated Stieltjes factors require the separate completion at spectral order 1 already developed in the source manuscript. No broad historical priority claim is attached to the kernel or this coefficient-extraction method.

Independent checks. The local subtraction above also produces an ordinary integrable function. For direct numerical evaluation, put $g_m(x) = \zeta(m, 1+x)$ and $R_{m,r}(x) = g_m(x) - \sum_{j=0}^{r-1} c_j(m)x^j$. Then

$$\begin{aligned} I_{kl} = &-\frac{1}{W-1} + \sum_{j=0}^{k-2}\frac{c_j(l)}{j-k+1} + \sum_{j=0}^{l-2}\frac{c_j(k)}{j-l+1} \\ &+ \int_0^1 \{g_k(x)g_l(x) + x^{-k}R_{l,k}(x) + x^{-l}R_{k,l}(x)\}\,dx. \end{aligned}$$

Both apparent singularities in the last integrand are removable. This representation gives a numerical left side independent of the Mikolás identity. For $k = l = 2$ it gives $1.61160883477664115747674903700975679 40\ldots$, and for $k = l = 3$ it gives $-6.91560891989003408335488060019362462 46\ldots$.

Source scope. The classical product integral should be credited to the source literature; see, for example, Shpot–Chaudhary–Paris, *Integrals of products of Hurwitz zeta functions and the Casimir effect in ϕ^4 field theories*, arXiv:1603.00722v2, Remark 1, and its cited Mikolás sources. That paper's Theorem 2 concerns the different ordinary integral of $\zeta(k, 1+x)^2$. The finite-part identity here is an explicit integer-resonance consequence with its coordinate correction proved. No claim of exhaustive historical novelty is made.

6 Spectral coordinates for derivative-containing polynomials

The preceding normal forms reduce higher digamma moments to differential polynomials containing at least one positive derivative. For this entire class, the following cancellation removes the convergence subtractions usually needed in a Mittag–Leffler expansion. All finite parts in this section use the original coordinate x at $x = 0$.

Put $\mathcal{R} = \mathbb{Q}[u_0, u_1, \ldots]$ and $\mathcal{J} = (u_1, u_2, \ldots) \subset \mathcal{R}$. For $P \in \mathcal{J}$, write

$$f_P(z) = P\big(\psi(z), \psi'(z), \ldots\big), \qquad f_P(-n+v) = \sum_{j=1}^{J} \frac{p_j(n)}{v^j} + O(1), \qquad r(n) = p_1(n).$$

Here J is any common upper bound for the pole orders; zero coefficients are understood. Let $\Delta a(n) = a(n) - a(n-1)$ for $n \geq 1$.

Theorem 6.1 (Unsubtracted expansion and finite-part coordinates)**.** *For every $P \in \mathcal{J}$,*

$$\boxed{f_P(z) = \sum_{n=0}^{\infty} \sum_{j=1}^{J} \frac{p_j(n)}{(z+n)^j}.} \tag{6.1}$$

The series is absolutely and locally uniformly convergent away from $0, -1, -2, \ldots$. For some fixed $A \geq 0$, its coefficients satisfy

$$p_j(n) = O\big(\log^A(n+2)\big), \qquad \Delta p_j(n) = O\big(n^{-1}\log^A(n+2)\big), \tag{6.2}$$

$$r(n) = O\big(n^{-1}\log^A(n+2)\big), \qquad \Delta r(n) = O\big(n^{-2}\log^A(n+2)\big). \tag{6.3}$$

Consequently the Dirichlet series

$$D_P(s) = \sum_{n=1}^{\infty} \frac{\Delta r(n)}{n^s} \tag{6.4}$$

is holomorphic for $\operatorname{Re} s > -1$, and

$$\boxed{\operatorname{FP}\int_0^1 f_P(x)\,\mathrm{d}x = \sum_{n=1}^{\infty} \sum_{j=2}^{J} \frac{\Delta p_j(n)}{(j-1)n^{j-1}} + D_P'(0).} \tag{6.5}$$

Every series in (6.5), including $D_P'(0) = -\sum_{n\geq 1} \Delta r(n)\log n$, converges absolutely.

Proof. The reflection formula and polygamma asymptotic expansions used below are standard Gamma-function identities; see [9, §§5.11,5.15]. Set

$$K(v) = -\pi\cot(\pi v), \qquad K'(v) = K(v)^2 + \pi^2.$$

For any real L, every $K^{(a)}$, $a \geq 1$, is a polynomial in K divisible by $K^2 + \pi^2$. Since each monomial of P contains a positive derivative, there is a polynomial T_L such that

$$P\big(L + K(v), K'(v), K''(v), \ldots\big) = \frac{\mathrm{d}}{\mathrm{d}v} T_L(K(v)). \tag{6.6}$$

Indeed, divide the left side by $K^2+\pi^2$, integrate the resulting polynomial in K, and use the displayed differential equation for K. Thus the residue of (6.6) at $v=0$ is identically zero.

Fix $0<\rho<1/2$. At the pole $-n$, reflection gives

$$\psi(-n+v)=K(v)+\log n+\eta_n(v),\qquad \eta_n(v)=\psi(n+1-v)-\log n.$$

Extend the subscript to real $t\ge 1$ in the rightmost formula. Uniformly on $|v|\le\rho$, any fixed finite collection of v derivatives satisfies

$$\partial_v^a\eta_t(v)=O(t^{-1}),\qquad \partial_t\partial_v^a\eta_t(v)=O(t^{-2})\quad (a\ge 0).$$

The first bound for $a=0$ includes the subtraction of $\log t$; the second includes its derivative. Coefficient extraction on $|v|=\rho$ therefore bounds every principal coefficient by a power of $\log t$, and its t derivative by $O(t^{-1}\log^A(t+2))$. For the residue, subtract (6.6) with $L=\log t$ before extracting the coefficient. Its residue is zero, while the difference is $O(t^{-1}\log^A(t+2))$ and its t derivative is $O(t^{-2}\log^A(t+2))$. The mean value theorem on $[n-1,n]$ gives (6.2)–(6.3); finitely many small n do not affect the estimates.

For completeness, these local calculations leave no undetermined entire function in (6.1). Use the circles $|w|=R_N=N+1/2$. They stay at least $1/2$ from every integer. In the right half-plane the usual asymptotics apply directly; in the left half-plane apply reflection to $1-w$. On these circles one obtains

$$|\psi(w)|\le C\log(R_N+2),\qquad |\psi^{(a)}(w)|\le C_a(R_N^{-1}+e^{-c|\operatorname{Im} w|})\quad (a\ge 1).$$

The exponential term follows from the positive derivatives of cotangent; pole avoidance makes the estimate uniform also when $|\operatorname{Im} w|$ is small. Hence

$$|f_P(w)|\le C\log^A(R_N+2)(R_N^{-1}+e^{-c|\operatorname{Im} w|}).$$

For fixed z,

$$\oint_{|w|=R_N}\frac{f_P(w)}{w-z}\,\mathrm{d}w=O\!\left(\frac{\log^A(R_N+2)}{R_N}\right)\longrightarrow 0,$$

because $\int_0^{2\pi}e^{-cR_N|\sin\theta|}\,\mathrm{d}\theta=O(R_N^{-1})$. The residue at $w=-n$ of the integrand is $-\sum_j p_j(n)/(z+n)^j$, and its residue at $w=z$ is $f_P(z)$. The residue theorem gives (6.1). Its absolute normal convergence follows from (6.2) and the extra factor n^{-1} in the estimate for $r(n)$.

Remove the finitely many principal terms at 0. The remaining series can be integrated uniformly on $[0,1]$. For $j\ge 2$, the pole at zero contributes $-p_j(0)/(j-1)$ to the finite part, and the other poles contribute

$$\sum_{n\ge 1}\frac{p_j(n)}{j-1}\bigl(n^{1-j}-(n+1)^{1-j}\bigr).$$

Summation by parts, with the vanishing boundary terms supplied by (6.2), gives the higher-pole sum in (6.5). The simple pole at zero has finite part zero. The remaining simple-pole terms give

$$\sum_{n\ge 1}r(n)\log\frac{n+1}{n}=-\sum_{n\ge 1}\Delta r(n)\log n=D_P'(0).$$

Here the boundary term $r(N)\log(N+1)$ tends to zero. The improved difference estimate proves the asserted holomorphy and all absolute convergence claims. □

Corollary 6.2 (Anchored antiderivative)**.** *Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$ and use the principal logarithm. For $z, z_0 \in \Omega$, the function*

$$F_P(z; z_0) = \sum_{n=0}^{\infty} r(n)(\mathrm{Log}(z+n) - \mathrm{Log}(z_0+n)) + \sum_{n=0}^{\infty} \sum_{j=2}^{J} \frac{p_j(n)}{j-1}((z_0+n)^{1-j} - (z+n)^{1-j}) \tag{6.7}$$

is locally normally convergent, satisfies $F_P(z_0; z_0) = 0$, and has $\partial_z F_P(z; z_0) = f_P(z)$. Its difference of logarithms must be retained for general complex anchors; it agrees with the principal logarithm of the ratio when z_0 is positive real.

Proof. On a fixed compact subset of $\Omega \times \Omega$, the logarithm difference is $O(n^{-1})$, while the inverse-power difference in the jth term is $O(n^{-j})$. The preceding coefficient bounds give normal convergence. Termwise differentiation and (6.1) finish the proof. □

6.1 A finite conversion to convergent multiple-zeta coordinates

Use the descending-index convention

$$\zeta(s_1, \ldots, s_\ell) = \sum_{n_1 > \cdots > n_\ell \geq 1} n_1^{-s_1} \cdots n_\ell^{-s_\ell}.$$

The depth is ℓ. Write $H_N(\alpha_1, \ldots, \alpha_\ell)$ for the same nested sum restricted to $N \geq n_1 > \cdots > n_\ell \geq 1$, and put $H_N(\varnothing) = 1$. A coordinate derivative differentiates one of the displayed arguments. For the statements below, coefficients belong to $\mathbb{Q}[\gamma, \zeta(2), \zeta(3), \ldots]$; ordinary zeta factors in this coefficient ring are not counted as additional depth.

Theorem 6.3 (Constructive depth bound)**.** *Let $d \geq 2$ be the largest number of factors in a monomial of P. Formula (6.5) has a finite expression in convergent multiple zeta values and first derivatives in one coordinate of multiple-zeta germs, all of depth at most $d-1$. Every germ can be chosen holomorphic for $\mathrm{Re}\, s > -1$, with s added to its first or second coordinate. The principal coefficients and this finite conversion are obtained by polynomial operations and finite stuffle products. If $d = 1$, the finite part in (6.5) is zero.*

Proof. Write $H_n^{(k)} = \sum_{a=1}^{n} a^{-k}$, with $H_n = H_n^{(1)}$. The exact local series is

$$\psi(-n+v) = -\frac{1}{v} + H_n - \gamma + \sum_{k \geq 1} (H_n^{(k+1)} + (-1)^{k+1}\zeta(k+1))v^k. \tag{6.8}$$

Differentiate and multiply to compute the finitely many required $p_j(n)$. Each term in a principal coefficient must use a singular factor, so $p_j(n)$ is a polynomial of harmonic degree at most $d-1$. Substitute $H_n^{(k)} = H_{n-1}^{(k)} + n^{-k}$ and subtract. Thus

$$\Delta p_j(n) = \sum_{e \geq 1} n^{-e} S_{j,e}(H_{n-1}, H_{n-1}^{(2)}, \ldots), \qquad \deg S_{j,e} \leq d-2, \tag{6.9}$$

with only finitely many nonzero terms. When $j \geq 2$, the outer exponent in (6.5) is $e+j-1 \geq 2$. Finite stuffle multiplication therefore converts these terms to convergent multiple zeta values of depth at most $d-1$. The same argument applies to the terms $e \geq 2$ in $D_P(s)$ and gives germs holomorphic for $\mathrm{Re}\, s > -1$.

The possible $e = 1$ terms require the residue cancellation, not an unjustified separation into divergent multiple-zeta series. Regard the residue as the finite polynomial $R(X_1, X_2, \ldots)$ obtained from (6.8). Setting $X_k = \zeta(k)$ for $k \geq 2$ turns that local series into $X_1 - \gamma + K(v)$. Equation (6.6) consequently gives the polynomial identity

$$R(X_1, \zeta(2), \zeta(3), \ldots) = 0 \quad \text{for every } X_1.$$

No algebraic independence of zeta values is being assumed here. The coefficient $S_{1,1}$ in (6.9) is $\partial_{X_1} R$. It belongs to the ideal generated by $X_k - \zeta(k)$, $k \geq 2$, and can be written as a sum of such factors times polynomials of degree at most $d - 3$; when $d = 2$ this coefficient vanishes. On substitution each factor has the convergent upper-tail expansion

$$H_{n-1}^{(k)} - \zeta(k) = -\sum_{a \geq n} a^{-k}, \qquad k \geq 2.$$

Expand the remaining polynomial by finite stuffle products. Each term then uses a composition α of length at most $d - 3$, and the exact conversion is

$$\boxed{\sum_{n \geq 1} \frac{(H_{n-1}^{(k)} - \zeta(k)) H_{n-1}(\alpha)}{n^{1+s}} = -\zeta(k, 1+s, \alpha) - \zeta(k+1+s, \alpha), \quad k \geq 2.} \tag{6.10}$$

Indeed, separate the upper-tail index $a \geq n$ into $a > n$ and $a = n$. The largest resulting depth is $2 + (d - 3) = d - 1$. Every prefix sum of real parts exceeds its prefix length when $\operatorname{Re} s > -1$, so both series on the right are absolutely convergent and holomorphic throughout that half-plane. The spectral parameter occupies the second coordinate in the first series and the first coordinate in the second. Differentiating at $s = 0$ completes the conversion of $D'_P(0)$.

If $d = 1$, P is a linear combination of positive derivatives of digamma. All its principal coefficients are constant in n, so (6.5) is zero. □

Corollary 6.4 (Depth of the Gamma normal-form remainder)**.** *For $m \geq 4$, write $\mathcal{N}(\mathcal{A}_m) = u_0^m + R_m$. The finite part of $R_m(\psi, \psi', \ldots)$ has a constructive expression in the preceding coordinates of depth at most $m - 3$.*

Proof. Every monomial of R_m contains a positive derivative. Its factor degree has the same parity as m, and its weight is m; after removal of u_0^m the factor degree is therefore at most $m - 2$. Apply Theorem 6.3. □

6.2 Two explicit consistency checks

For $P = u_1^2$, the nonzero principal coefficients are

$$p_4(n) = 1, \qquad p_2(n) = 2\big(H_n^{(2)} + \zeta(2)\big), \qquad r(n) = 4\big(H_n^{(3)} - \zeta(3)\big).$$

The general formula immediately yields

$$\mathrm{FP} \int_0^1 \psi'(x)^2 \, \mathrm{d}x = 2\zeta(3) + 4\zeta'(3). \tag{6.11}$$

This is also a useful numerical check because all subtractions at the origin are explicit.

For $P = u_0 u_1^2$, the residue difference is

$$\Delta r(n) = \frac{4(H_{n-1}^{(3)} - \zeta(3))}{n} + \frac{2(H_{n-1}^{(2)} + \zeta(2))}{n^2} + \frac{4(H_{n-1} - \gamma)}{n^3}.$$

The upper-tail conversion gives the fully convergent expression

$$\begin{aligned} D_P(s) = & -4\zeta(3,s+1) - 4\zeta(s+4) + 2\zeta(s+2,2) + 2\zeta(2)\zeta(s+2) \\ & + 4\zeta(s+3,1) - 4\gamma\zeta(s+3), \qquad \operatorname{Re} s > -1. \end{aligned} \tag{6.12}$$

In particular,

$$\operatorname{FP}\int_0^1 \psi(x)\psi'(x)^2\,\mathrm{d}x = \frac{35}{6}\zeta(4) - 2\gamma\zeta(3) + D_P'(0). \tag{6.13}$$

The term $\zeta(3, s+1)$ illustrates why the spectral derivative may move to an interior coordinate. Separately writing $\zeta(s+1,3) - \zeta(3)\zeta(s+1)$ at $s = 0$ would hide the cancellation that the upper-tail argument makes explicit.

7 An explicit sixth digamma moment and higher-weight consequences

We now apply the two independent reductions: the scalar Gamma bridge isolates the strict-depth Laurent slopes, while the differential normal form leaves a small number of derivative-containing integrals. At weight six only one genuinely mixed coordinate remains after the quadratic polygamma formula has been used. We specify that coordinate completely, including ordinary convergent series and a finite multiple-zeta representation.

7.1 The mixed sixth-weight coordinate

At $x = -n + v$ write

$$\psi(x) = -v^{-1} + b_n + c_n v + d_n v^2 + e_n v^3 + f_n v^4 + O(v^5),$$

where, only within this subsection,

$$\begin{gathered} b_n = H_n - \gamma, \quad c_n = H_n^{(2)} + \zeta(2), \quad d_n = H_n^{(3)} - \zeta(3), \\ e_n = H_n^{(4)} + \zeta(4), \quad f_n = H_n^{(5)} - \zeta(5). \end{gathered} \tag{7.1}$$

These local c_n are unrelated to the reciprocal-Gamma coefficients c_j of (4.11). Direct multiplication gives the principal parts of $f = \psi^2\psi'^2$:

j	$p_j(n)$ in $f(-n+v) = \sum_{j=1}^{6} p_j(n)v^{-j} + O(1)$
6	1
5	$-2b_n$
4	b_n^2
3	$-2b_nc_n + 2d_n$
2	$2b_n^2c_n - 2c_n^2 - 6b_nd_n + 4e_n$
1	$4b_n^2d_n + 2b_nc_n^2 - 6c_nd_n - 10b_ne_n + 6f_n$

Let $r_n = p_1(n)$ and define

$$\mathcal{D}(s) = \sum_{n\ge 1}(r_n - r_{n-1})n^{-s}, \qquad \Omega_5 = \mathcal{D}'(0). \tag{7.2}$$

By the residue theorem of Section 6, $\mathcal{D}$ is holomorphic for $\operatorname{Re} s > -1$. In particular these are ordinary absolutely convergent identities:

$$\boxed{\Omega_5 = -\sum_{n\ge 1}(r_n - r_{n-1})\log n = \sum_{n\ge 1} r_n \log\left(1 + \frac{1}{n}\right).} \tag{7.3}$$

The last equality is summation by parts; its boundary term $r_N \log(N+1)$ tends to zero. The subscript five indicates the residual weight of the moment calculation, not a Stieltjes index or a zeta derivative order.

For a short direct check, set $b = b_{n-1}, c = c_{n-1}, d = d_{n-1}, e = e_{n-1}$. Replacing the harmonic cutoff $n-1$ by n in the residue polynomial gives exactly

$$r_n - r_{n-1} = \frac{8bd + 2c^2 - 10e}{n} + \frac{4bc - 2d}{n^2} + \frac{4b^2 - 2c}{n^3} - \frac{4}{n^5}. \tag{7.4}$$

Its apparent $1/n$ term has a vanishing frozen coefficient. Indeed, with $T_j = \zeta(j) - H_{n-1}^{(j)}$,

$$8bd + 2c^2 - 10e = -8bT_3 - 8\zeta(2)T_2 + 2T_2^2 + 10T_4, \tag{7.5}$$

where $8\zeta(2)^2 = 20\zeta(4)$ was used. This also directly proves the convergence asserted in (7.2). Telescoping without logarithms gives the useful exact normalization

$$\mathcal{D}(0) = 4\gamma^2\zeta(3) - 5\gamma\zeta(4) - 6\zeta(2)\zeta(3) + 6\zeta(5). \tag{7.6}$$

The finite multiple-zeta representation can be stated compactly as $\mathcal{D}(s) = \sum_{\boldsymbol{a}} q_{\boldsymbol{a}} \zeta(s + a_1, a_2, \ldots, a_d)$, with the following complete table. The entries with first index one are combined before continuation or differentiation; an individually convergent alternative is given in Appendix A.2.

$\boldsymbol{a}$	$q_{\boldsymbol{a}}$	$\boldsymbol{a}$	$q_{\boldsymbol{a}}$
(1)	$8\gamma\zeta(3) + 2\zeta(2)^2 - 10\zeta(4)$	$(2,2)$	-4γ
$(1,1)$	$-8\zeta(3)$	$(2,2,1)$	4
$(1,1,3)$	8	$(2,3)$	2
$(1,2)$	$4\zeta(2)$	(3)	$4\gamma^2 - 2\zeta(2)$
$(1,2,2)$	4	$(3,1)$	-8γ
$(1,3)$	-8γ	$(3,1,1)$	8
$(1,3,1)$	8	$(3,2)$	2
(2)	$-4\gamma\zeta(2) + 2\zeta(3)$	(5)	-4
$(2,1)$	$4\zeta(2)$	$(2,1,2)$	4

To prove the table, expand (7.4) and use only $H_N^{(a)} H_N^{(b)} = H_N(a,b) + H_N(b,a) + H_N^{(a+b)}$. No numerical recognition or conjectural relation is needed.

Proposition 7.1 (The complete mixed integral)**.** *With* Ω_5 *defined by* (7.2)*,*

$$\boxed{\mathrm{FP}\int_0^1 \psi(x)^2\psi'(x)^2\,\mathrm{d}x = \Omega_5 + 2\gamma^2\zeta(3) - \frac{35}{3}\gamma\zeta(4) + \frac{31}{3}\zeta(2)\zeta(3) - \frac{19}{3}\zeta(5).} \tag{7.7}$$

Proof. The nonresidue part of the general finite-part formula is $J = \sum_{n\geq 1}\sum_{j=2}^{6}(p_j(n) - p_j(n-1))/((j-1)n^{j-1})$. The pole table reduces its entire summand to

$$j_n = \frac{4bc - 6d}{n^2} + \frac{2b^2 - 3c}{n^3} - \frac{7b}{3n^4} - \frac{13}{6n^5}.$$

Stuffle expansion expresses its sum in convergent multiple zeta values of weight at most five. The needed reductions are

$$\zeta(2,1) = \zeta(3), \quad \zeta(3,1) = \tfrac{1}{4}\zeta(4), \quad \zeta(2,2) = \tfrac{3}{4}\zeta(4),$$

$$\zeta(4,1) = 2\zeta(5) - \zeta(2)\zeta(3),$$
$$\zeta(3,2) = 3\zeta(2)\zeta(3) - \tfrac{11}{2}\zeta(5),$$
$$\zeta(2,3) = \tfrac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3),$$

together with the dualities $\zeta(3,1,1) = \zeta(4,1)$, $\zeta(2,2,1) = \zeta(3,2)$, and $\zeta(2,1,2) = \zeta(2,3)$. These classical low-weight identities can be obtained by the iterated-integral substitution $t \mapsto 1-t$ and the shuffle/stuffle product rules. For example the weight-five double sum formula, the stuffle of $\zeta(2)\zeta(3)$, and its shuffle give respectively

$$\zeta(2,3) + \zeta(3,2) + \zeta(4,1) = \zeta(5),$$
$$\zeta(2)\zeta(3) = \zeta(2,3) + \zeta(3,2) + \zeta(5) = \zeta(2,3) + 3\zeta(3,2) + 6\zeta(4,1),$$

which are a nonsingular linear system for the displayed evaluations. Substitution in $\sum j_n$ gives $J = 2\gamma^2\zeta(3) - 35\gamma\zeta(4)/3 + 31\zeta(2)\zeta(3)/3 - 19\zeta(5)/3$. The residue contribution is $\mathcal{D}'(0) = \Omega_5$, proving (7.7). □

7.2 The sixth power with no unnamed mixed integral

Return to $c_j = [u^j]\Gamma(1+u)^{-1}$. The required coefficients are

$$\begin{aligned} c_2 &= \frac{\gamma^2 - \zeta(2)}{2}, \\ c_3 &= \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{6}, \\ c_4 &= \frac{\gamma^4 - 6\gamma^2\zeta(2) + 3\zeta(2)^2 + 8\gamma\zeta(3) - 6\zeta(4)}{24}. \end{aligned} \tag{7.8}$$

Theorem 7.2 (The sixth digamma moment)**.** *In the original coordinate finite-part convention,*

$$\boxed{\begin{aligned} \mathrm{FP}\int_0^1 \psi(x)^6\,\mathrm{d}x = {} & 15\Omega_5 - 720\lambda_5 + 720\gamma\lambda_4 - 720c_2\lambda_3 + 720c_3\lambda_2 + 720c_4\gamma_1 \\ & - 60\gamma^3\zeta(2) + 180\gamma^2\zeta(3) - 70\gamma\zeta(4) \\ & - 55\zeta(2)\zeta(3) + 175\zeta(5) + 48\zeta'(5) - 135\zeta(6). \end{aligned}} \tag{7.9}$$

All constants on the right have specified ordinary-series, Laurent, or standard special-function definitions. In particular, this formula makes no assertion that Ω_5 reduces to a smaller constant alphabet.

Proof. The exact primitive in the sixth-weight normal decomposition is

$$H_6 = -3u_0^5 + 20u_0^3u_1 - 15u_0^2u_2 - 15u_0u_1^2 + 6u_0u_3 + 9u_1u_2 - u_4. \tag{7.10}$$

Direct differentiation gives $\mathcal{A}_6 = u_0^6 - 15u_0^2u_1^2 + u_2^2 + DH_6$. The Laurent expansion (4.3) gives

$$\Delta H_6 = 60\gamma^3\zeta(2) - 150\gamma^2\zeta(3) - 105\gamma\zeta(4) + 210\zeta(2)\zeta(3) - 242\zeta(5). \tag{7.11}$$

The scalar bridge is

$$\mathcal{L}_6 = -135\zeta(6) - 720\{c_4\lambda_1 - c_3\lambda_2 + c_2\lambda_3 - \gamma\lambda_4 + \lambda_5\}.$$

The Hurwitz-product formula gives $\mathrm{FP}\int \psi''(x)^2\,\mathrm{d}x = -28\zeta(5) - 48\zeta'(5)$. Therefore

$$Q_6 = \mathcal{L}_6 + 15\,\mathrm{FP}\int_0^1 \psi^2\psi'^2\,\mathrm{d}x - \mathrm{FP}\int_0^1 \psi''^2\,\mathrm{d}x - \Delta H_6.$$

Substitute (7.7), (7.11), and $\lambda_1 = -\gamma_1$; collecting rational coefficients proves (7.9). □

As numerical reference values, independently evaluated quantities are

$$\Omega_5 = -10.139865313210457474013040008351766 65\ldots,$$
$$\mathrm{FP}\int_0^1 \psi^2\psi'^2\,\mathrm{d}x = -2.762494287617278704462082548609619 92\ldots,$$
$$Q_6 = -64.955112035383345752175350958185790 02\ldots.$$

The mixed-identity residual was below $2\cdot 10^{-99}$ in the 115-digit run. An end-to-end check of (7.9), using independent Mellin integrals for all five λ_d, had residual below 10^{-87} at 90 working digits. These are reproducible floating-point diagnostics, not certified error enclosures.

7.3 Seventh, eighth, and arbitrary weights

The next two normal forms in (4.10) give

$$Q_7 = \mathcal{L}_7 - \Delta H_7 + 35\,\mathrm{FP}\int_0^1 \psi^3\psi'^2\,\mathrm{d}x - 7\,\mathrm{FP}\int_0^1 \psi\psi''^2\,\mathrm{d}x, \tag{7.12}$$

$$Q_8 = \mathcal{L}_8 - \Delta H_8 + 70\,\mathrm{FP}\int_0^1 \psi^4\psi'^2\,\mathrm{d}x - 28\,\mathrm{FP}\int_0^1 \psi^2\psi''^2\,\mathrm{d}x + 7\,\mathrm{FP}\int_0^1 \psi'^4\,\mathrm{d}x + \mathrm{FP}\int_0^1 \psi'''^2\,\mathrm{d}x. \tag{7.13}$$

These are useful compact reductions rather than claims of minimal arithmetic bases. Every remaining integral is evaluated by the general pole-coefficient procedure, and the last square is already the explicit quadratic Hurwitz value. The endpoint polynomials $\Delta H_7, \Delta H_8$ appear in Appendix A.1; the data file retains all exact coefficients without a choice of even-zeta simplification.

At general weight m, the closed coefficient sectors determine every term $\psi^{m-2r-2}(\psi^{(r)})^2$ and every term containing only ψ and ψ'. The remaining functional monomials can be generated finitely. Their residue differences have finite harmonic word expansions, and their antiderivatives follow from the normally convergent primitive theorem. This supplies a systematic way to produce exact identities and isolate the next arithmetic question at each weight. It does not turn formal dimension into a theorem about independence of numerical constants.

8 Source audit, proposed corrections, and further research

8.1 What the audit establishes

The repository was inspected at the revision on the title page. All six incoming ZIPs were unpacked with a path-safety check; their inventories, claim/status files, and relevant sections were read. The accompanying `incoming_audit_summary.json` records the archive names, SHA-256 digests, already covered results, and open questions. This was a targeted mathematical audit of the claims used here, not a line-by-line certification of every theorem in the large source corpus.

The source claims used most directly are the exact even-tail Bernoulli coordinates, the first-q conjecture, the Gamma-quotient ordinary integral, the fifth-moment baseline, and the recorded Family–2 example. The tail coefficient/operator conversion was checked independently; the new proof includes the column reversal and the omitted lower-index-zero convention. The Gamma bridge was reproved with its Laurent normalization before it was used in the sixth moment. The Family–2 equation was compared to its original right-hand side, not merely to a numerically similar simplification.

No new false theorem in the inherited material was established by this audit. Some incoming reports already correct signs, Stieltjes indexing, or expressibility claims in external papers. Those corrections are identified as inherited in the audit record; they are not counted as new findings here.

8.2 Concrete proposed source changes

1. **Upgrade the Family–2 status.** In the canonical `chapters/04-depth.tex`, the equation labeled `double:eq:fam2-example` can be marked proved. Retain its original identity for continuity, add (2.10) as its simplified form, and cite the two exact component evaluations (2.8)–(2.9). The discovery by integer relation remains a useful historical note, but it is no longer the justification for that equation.

2. **Refine the optimal-prefix conjecture.** In the incoming canonical fifth-moment report, retain `conj:firstq` for general unequal even orders. Record $p = q - 2$ as a theorem with determinant (3.9); retain $p = q$ as the already proved source case and general gaps $q - p \geq 4$ as open. This is an infinite-family resolution, not a complete proof of the original conjecture.

3. **Attach the sixth calculation to the higher-moment agenda.** The new result (7.9) completes a specific sixth-power calculation in explicit convergent zeta-derivative coordinates. Its remaining arithmetic question is the reduction of Ω_5. The generic all-power algorithm and the fifth calculation were already present and should retain their earlier attribution.

4. **Cite the exact antecedents for two reusable methods.** The functional-monomial direct sum should cite [6, Proposition 4.10]; the bicolored parity implementation should cite Nakamura and Panzer [4, 5]. The latter paper's footnote 16 identifies a transposed product in Nakamura's printed equation (1.3). Our proof does not import that formula without rederivation. This is an attribution and formula-use clarification, not an assertion that the supplied manuscript copied the misprint.

5. **Preserve the precise meanings of "finite part" and "rank".** When importing the new results, keep the original-coordinate definition (1.6), the local correction in the Hurwitz product section, and the formal coefficient-space qualifiers. A raw Laurent constant of the spectral kernel generally differs from the coordinate integral; $\mathcal{N}$ is not multiplicative; and a formal rank computation does not prove arithmetic independence. These are explicit integration requirements for this article, not newly alleged errors in every prior use of those terms.

8.3 Research questions with concrete mathematical targets

The following agenda separates exact next calculations from genuine structural questions. An answer should supply a proof, a fully stated functional identity, or an exact algebraic certificate. A large precision search or a numerical rank alone would not settle the corresponding question.

1. The first unresolved even-tail gap. Prove or disprove the optimal first-q conjecture for $q = p + 4$, uniformly in p. The adjacent-order proof uses one free formal coefficient and a sign-definite scalar recurrence. The next gap produces a larger initial-data space. The concrete target is a nonzero determinant formula for the associated finite block, perhaps through a matrix recurrence whose relevant minors have a controlled sign. This would test whether the adjacent argument extends or depends essentially on second order.

2. A determinant formula for arbitrary unequal orders. For $p = 2a, q = 2b$, seek a closed factorization or a finite positive coefficient formula for the late Bernoulli matrix determinant. Any proposed expression must retain the source's omitted binomial terms and the centered cutoff. A proof should also give the inverse map, thereby producing ordinary harmonic-tail identities in addition to a rank statement.

3. Arithmetic structure of the adjacent determinant sequence. The rational numbers c_n have both a Bernoulli recurrence and the central-binomial representation (3.29). Determine an exact denominator formula, a useful integer renormalization, or a finite convolution identity for this sequence. The target is an arithmetic identity or recurrence with transparent integrality, rather than an asymptotic estimate.

4. The next full Gamma coefficient sector. Find the generating function for all functional monomials containing u_2^2 and arbitrary powers of u_0, u_1, including any terms coupled to u_2^3 by factor-degree parity. The ideal argument used for the complete u_0, u_1 sector suggests working modulo a precisely stated higher-derivative monomial ideal. Its preservation by the normal projection must be proved from the reduction rule; stability under D by itself is insufficient.

5. Higher sectors and a global normal-form generating function. The centered generator contains only even derivative indices before normal reduction. Can that symmetry be turned into a closed multivariable generating function for $\mathcal{N}(\mathcal{A}_m)$, beyond the square and digamma–trigamma sectors? An exact coefficient-extraction formula indexed by partitions would be useful even without a short elementary expression.

6. Arithmetic reduction of the mixed sixth coordinate. Determine whether Ω_5 can be expressed in a prescribed algebra generated by the height-one λ_d, ordinary zeta derivatives, and lower mixed coordinates already appearing in the fifth moment. Appendix A.2 supplies a finite list of absolutely convergent MZV germs of depth at most three. This makes the question precise enough for a symbolic functional relation search. Failure of a finite search would not prove nonmembership or independence.

7. Complex-order functional relations before differentiation. The preceding question is about derivatives in zeta orders, not only values at integral orders. A parity identity known only for integer indices cannot be differentiated in an index without an analytic extension. Seek functional equations for the required complex-order germs, prove their continuation and local regularity, and only then differentiate. This is a specific way to avoid an unjustified "differentiate the MZV reduction" step.

8. Minimal spectral depth of higher moments. The normal-form and residue theorems give an upper bound on the depth needed for the derivative-containing remainder of Q_m. Determine when that bound is attained after all proven functional relations are used. First complete explicit seventh- and eighth-weight coordinate tables, then look for uniform cancellations. Distinguish the dimension of differential-polynomial normal forms, the rank of their harmonic residue maps, and the dimension of their actual period values; these are different problems.

9. Generalized Stieltjes products at the missing derivative edge. The finite generator in (5.8) treats every pair of positive argument-derivative orders. Extend its manifestly

holomorphic coefficient form to cases with one or both derivative orders zero, and prove compatibility with the incoming Stieltjes-collision formulas. The extra spectral poles at order one must be subtracted jointly; blindly setting an integer parameter to one in the positive-order formula does not establish this extension.

10. Exact cyclotomic identities in the complementary parity sector. The bicolored theorem computes real parts in odd weight and imaginary parts in even weight. The Gaussian S_6 and revised S_8 questions are not resolved by that mechanism. Seek a genuinely additional relation: a proven depth-three reduction, an integral symmetry, or a specialization of a functional equation with a larger argument set. The precise definitions and corrected candidates in the current source should be retained; already rejected coefficient lists should not be recycled as conjectures.

11. Antiderivatives and a specified function class. The spectral theorem provides normally convergent anchored primitives for every derivative-containing differential polynomial. Which of these primitives reduce to a specified finite class of Gamma, Barnes, polylogarithm, and zeta-order-derivative functions? A useful first target is the primitive of $\psi^2\psi'^2$. Any nonexistence claim must name the allowed class and prove an obstruction within it; an unspecified notion of "closed form" does not define a mathematical problem.

12. Exact proof certificates for repository integration. Turn the polynomial reductions, low-weight shuffle identities, and finite Bernoulli matrix conjugacies into machine-readable exact certificates. The current scripts already separate rational identities from numerical diagnostics. A next stage could check those certificates with an independent implementation or a proof assistant, while leaving analytic continuation and contour estimates as explicit lemmas with their assumptions stated.

The most immediate next proof target is the $q-p=4$ tail determinant. The most immediate next arithmetic target is the convergent depth-three expression for Ω_5. Both have precise finite input data in this article, and both would produce exact identities of direct use to the existing manuscript.

A Exact endpoint and convergent-coordinate tables

A.1 Endpoint corrections through weight eight

For the unique primitives $H_m = \mathcal{H}(\mathcal{A}_m)$ in (4.1), the endpoint functional is

$$\begin{aligned}
\Delta H_2 &= 0,\\
\Delta H_3 &= -3\zeta(2),\\
\Delta H_4 &= 12\gamma\zeta(2) - 10\zeta(3),\\
\Delta H_5 &= -30\gamma^2\zeta(2) + 50\gamma\zeta(3) + \tfrac{35}{2}\zeta(4),\\
\Delta H_6 &= 60\gamma^3\zeta(2) - 150\gamma^2\zeta(3) - 105\gamma\zeta(4) + 210\zeta(2)\zeta(3) - 242\zeta(5),\\
\Delta H_7 &= -105\gamma^4\zeta(2) + 350\gamma^3\zeta(3) + \tfrac{735}{2}\gamma^2\zeta(4)\\
&\quad - 1470\gamma\zeta(2)\zeta(3) + 1694\gamma\zeta(5) + 469\zeta(3)^2 - \tfrac{1085}{2}\zeta(6),\\
\Delta H_8 &= 168\gamma^5\zeta(2) - 700\gamma^4\zeta(3) - 980\gamma^3\zeta(4)\\
&\quad + 5880\gamma^2\zeta(2)\zeta(3) - 6776\gamma^2\zeta(5) - 3752\gamma\zeta(3)^2 + 4340\gamma\zeta(6)\\
&\quad + 7840\zeta(2)\zeta(5) - 1862\zeta(3)\zeta(4) - 11824\zeta(7).
\end{aligned}$$

They use only the elementary simplifications $\zeta(2)^2 = 5\zeta(4)/2$, $\zeta(2)^3 = 35\zeta(6)/8$, and $\zeta(2)\zeta(4) = 7\zeta(6)/4$. The data file `inverse_gamma_normal_forms.json` retains the original independent-symbol expressions as well as each exact primitive and normal form. Repeating the finite Laurent coefficient extraction proves every entry of this table.

A.2 A fully convergent 17-term representation of $\mathcal{D}(s)$

The first-index-one table in Section 7 is convenient for formal harmonic-word generation, but individual entries there may have spectral poles. The following alternative avoids them entirely. The function $\mathcal{D}(s)$ is the sum of each function in the left column multiplied by its coefficient in the right column:

Individually convergent MZV germ	Coefficient
$\zeta(2+s)$	$2\zeta(3) - 4\gamma\zeta(2)$
$\zeta(3+s)$	$4\gamma^2 - 8\zeta(2)$
$\zeta(4+s)$	8γ
$\zeta(5+s)$	6
$\zeta(2+s,1)$	$4\zeta(2)$
$\zeta(2+s,2)$	-4γ
$\zeta(2+s,3)$	2
$\zeta(3+s,1)$	-8γ
$\zeta(4+s,1)$	-8
$\zeta(2+s,1,2)$	4
$\zeta(2+s,2,1)$	4
$\zeta(3+s,1,1)$	8
$\zeta(2,1+s)$	$-6\zeta(2)$
$\zeta(3,1+s)$	8γ
$\zeta(4,1+s)$	10
$\zeta(2,1+s,2)$	-2
$\zeta(3,1+s,1)$	-8

Every entry converges absolutely and defines a holomorphic function on $\operatorname{Re} s > -1$. Therefore Ω_5 is also the sum of the *individual ordinary derivatives at zero* of these 17 germs, with the same coefficients. Derivatives in the second coordinate must remain second-coordinate derivatives.

For an exact derivation, use the elementary identity, valid for $k \geq 2$ and a positive-index word $\boldsymbol{a}$,

$$\sum_{n\geq 1} \frac{(H_{n-1}^{(k)} - \zeta(k))H_{n-1}(\boldsymbol{a})}{n^{1+s}} = -\zeta(k, 1+s, \boldsymbol{a}) - \zeta(k+1+s, \boldsymbol{a}). \tag{A.1}$$

It follows by writing the negative tail as $-\sum_{j\geq n} j^{-k}$ and partitioning $j > n$ from $j = n$. All terms are absolutely summable in the stated half-plane; any additional inner unit indices contribute at most logarithms. The coefficient of $1/n$ in (7.4) factors exactly as

$$8(H_{n-1} - \gamma)(H_{n-1}^{(3)} - \zeta(3)) + 2(H_{n-1}^{(2)} - \zeta(2))(H_{n-1}^{(2)} + 3\zeta(2)) - 10(H_{n-1}^{(4)} - \zeta(4)).$$

Apply (A.1) to these three terms and retain the first-index-at-least-two rows of the original table. Two $\zeta(3+s,2)$ terms cancel; collecting the single zeta terms gives precisely the 17

rows above. This establishes the alternative without analytic regularization of any individual summand.

The same residue calculation gives an exact additional continuation statement. Its first two asymptotic coefficients are $r_n - r_{n-1} = 8\zeta(2)(\log n - 1)/n^2 + O(\log^2(n+2)/n^3)$, obtained by substituting the ordinary harmonic expansions in (7.4). Consequently

$$\mathcal{D}(s) + 8\zeta(2)\{\zeta'(s+2) + \zeta(s+2)\}$$

is holomorphic for $\operatorname{Re} s > -2$. In particular the exact principal part of $\mathcal{D}$ at $s = -1$ is

$$\frac{8\zeta(2)}{(s+1)^2} - \frac{8\zeta(2)}{s+1}.$$

The asymptotic estimate here is used only to prove that spectral identity and its pole coefficients.

A.3 A convergent logarithmic antiderivative for the mixed term

For $a > 0$ and $\operatorname{Re} z > 0$, the pole table in Section 7 gives the anchored primitive

$$\begin{aligned}\mathcal{P}_6(z;a) = \sum_{n\ge 0}\Bigg\{ & r_n(\operatorname{Log}(n+z) - \log(n+a)) \\ & + \sum_{j=2}^{6} \frac{p_j(n)}{1-j}\big((n+z)^{1-j} - (n+a)^{1-j}\big)\Bigg\}.\end{aligned} \tag{A.2}$$

It converges normally on compact subsets of the right half-plane, vanishes at $z = a$, and satisfies $\partial_z \mathcal{P}_6(z;a) = \psi(z)^2\psi'(z)^2$. Thus the mixed sixth coordinate has a fully specified antiderivative as well as its endpoint finite-part value. The general primitive theorem supplies the same construction for all the derivative terms in the seventh, eighth, and higher moment reductions.

B Reproducibility and the limits of computation

The package separates proof, exact verification, and floating-point diagnostics. The proofs of the infinite families appear in the article. Exact checks compare independent finite constructions or verify finite polynomial identities. Numerical residuals test the analytic formulas at selected instances; they are not interval certificates and do not prove arbitrary-order assertions.

Verification family	Completed evidence
Bicolored parity	49 exact partial-fraction checks; 48 independent Mellin-integral parity comparisons, plus explicit specializations and the original Family–2 right-hand side. The largest parity residual was $5.94 \cdot 10^{-50}$ at 50 working digits.
Adjacent tails	127 exact assertions, including matrix/operator conjugacy and determinants through $b = 16$, recurrence signs through c_{40}, and source-coordinate cancellation. Eight numerical diagnostics had maximum scaled residual $1.62 \cdot 10^{-77}$.
Gamma sectors	Independent Bell-polynomial and normal-form construction through weight 16; 49 square coefficients, 17 full sectors, and the hyperbolic-sine-integral generator agree exactly.
Hurwitz products	Four independent local-subtraction quadratures and one logarithmic order-derivative quadrature, all at 85 working digits; absolute residuals were below $7 \cdot 10^{-85}$.
Spectral polynomials	Independent principal-part checks, frozen-residue cancellation, the exact mixed residue difference, and a separately regularized trigamma-square quadrature.
Mixed sixth coordinate	Six principal parts, a Laurent constant, two formal harmonic-word tables, 24 finite-harmonic checks, 54 finite tail-ordering checks, and the exact weight-five reduction. The 115-digit mixed identity residual was $1.76 \cdot 10^{-99}$.
Full sixth moment	Independent local-series quadrature for Q_6 and Mellin integrals for all five strict-depth slopes. The 90-digit end-to-end residual was $9.90 \cdot 10^{-88}$.

The scripts require Python 3, SymPy, and mpmath. The actual run used SymPy 1.14.0 and mpmath 1.3.0. Exact arithmetic uses rational numbers or independent symbolic constants. In the numerical tail calculations, the finite sum and the asymptotic Hurwitz tail are evaluated with separately specified cutoffs and working precision; the JSON records retain those parameters and the observed discrepancies.

The package includes:

- the compiled PDF, one self-contained TeX source, and modular TeX sources with stable theorem labels;
- exact Gamma normal-form and mixed-coordinate data;
- independent verification scripts and their completed JSON records;
- the pinned source/archive audit, a claim ledger, and proposed integration changes;
- a build command, dependency notes, and a package checksum manifest.

No original incoming report is silently replaced. No source conjecture outside the explicitly proved cases is marked solved. A suggested integration is to add the article as a new incoming research report, then promote the two proved source claims using the precise statements in Section 8. The source revision and archive digests make that comparison reproducible even if the repository subsequently changes.

References

[1] ProveIt project, *Polylogarithms manuscript*, canonical modular manuscript, `Analysis/Polylogarithms/docs/manuscript`, repository revision `f7e35be357654198c1546fd40f363694c5bcc2cc`. https://github.com/VladimirReshetnikov/ProveIt. The source equation `double:eq:fam2-example` is in `chapters/04-depth.tex`; the coincident Hurwitz kernel is in `chapters/07-shifted-stieltjes-correlations.tex`.

[2] ProveIt project, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge*, incoming report in `ProveIt_Canonical_Digamma_Fifth_Moment_2026-10-11.zip`, at the same pinned revision. Sections 3, 7, and 9 contain the Gamma bridge, exact even-tail coordinates, and Conjecture `conj:firstq`. Archive SHA-256: `3c2fa00434f0366089b819bad77c5a176443594ed1fe6dd0731c877bc74d01a7`.

[3] ProveIt project, *Higher Digamma Powers and Ordered Gamma Sums*, incoming report in `ProveIt_Higher_Digamma_Powers_2026-10-11.zip`, at the same pinned revision. The all-power algorithm, fifth moment, and normalized primitives are prior baseline results.

[4] T. Nakamura, *A simple proof of the functional relation for the Lerch type Tornheim double zeta function*, Tokyo Journal of Mathematics **35** (2012), 333–337. https://arxiv.org/abs/1012.1144. See Proposition 1.2, read with the product correction recorded by Panzer in footnote 16 of the next reference.

[5] E. Panzer, *The parity theorem for multiple polylogarithms*, Journal of Number Theory **172** (2017), 93–113. https://arxiv.org/abs/1512.04482. See Theorem 1.3, equation (3.2), and footnote 16.

[6] L. Guo, G. Regensburger, and M. Rosenkranz, *On integro-differential algebras*, Journal of Pure and Applied Algebra **218** (2014), 456–473. https://doi.org/10.1016/j.jpaa.2013.06.015; https://arxiv.org/abs/1212.0266. Proposition 4.10 gives the functional-monomial direct-sum decomposition.

[7] M. Rosenkranz and G. Regensburger, *Integro-differential polynomials and operators*, Proceedings of ISSAC 2008, 261–268. The reduction rule is given in Lemma 5; the later treatment [6] supplies the direct-sum formulation used here.

[8] M. A. Shpot, M. P. Chaudhary, and R. B. Paris, *Integrals of products of Hurwitz zeta functions and the Casimir effect in ϕ^4 field theories*, arXiv:1603.00722v2 (2016). https://arxiv.org/abs/1603.00722. Remark 1 states the classical Mikolás product integral. Theorem 2 concerns the shifted, ordinary product integral, which is distinct from the coordinate finite part evaluated in this article.

[9] NIST Digital Library of Mathematical Functions, https://dlmf.nist.gov/. Relevant formulas and sections: 5.5.4 (digamma reflection), 5.7 (Gamma and digamma series), 5.11.2 and 5.15.9 (digamma and polygamma asymptotics), 24.8.1 (Bernoulli Fourier series), and 25.11 (Hurwitz zeta). Accessed 11 October 2026.